

IT 320 Course Project  
Semester-2, 1447H



## Software Product Release

<Qinwan - قِنْوَان>

Sprint 0– Phase 2

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# 1 Chapter 1: Introduction

Technology has quietly reshaped how people invest. Today, someone in Riyadh can own a fraction of a property in Doha, fund a startup in London, or trade stocks in New York , all from their phone. Yet one sector has remained stubbornly out of reach for the everyday person: agriculture. In Saudi Arabia, date palm farming is not just an industry, it is a cultural identity, a national heritage, and an increasingly strategic pillar of the Kingdom's Vision 2030. With over 37.6 million date palms and a national production of approximately 1.9 million tons in 2024 [1], the scale of this sector is undeniable. However the ability to participate in it financially has always required something most people simply do not have : land, large capital, or the right connections. Conversely, many farmers who own productive land and possess the necessary skills cannot find interested individuals willing to invest in their farms or purchase their future harvests.

The gap between those who want to invest in this thriving sector and those who own the land to make it possible has persisted for years, not because there is no interest on either side, but because no structured, accessible bridge between them has ever existed.

Introducing “Qinwan”, web-based platform that enables simple and transparent investment in date palm farms across Saudi Arabia. The platform bridges the gap between individual investors and farm owners, allowing users to discover and invest in specific land of palm trees, track their growth, and choose flexible returns from their harvest all through a secure and user-friendly digital interface.

At the local level, a technological solution in this domain plays a direct role in supporting farmers and revitalizing agricultural activity across the Kingdom. By increasing visibility of farms and enabling structured investment participation, farmers gain improved access to financial resources, allowing them to sustain operations, enhance productivity, and plan for future agricultural cycles more effectively. This contributes to the continuity and growth of farming activities, particularly in rural regions where agriculture remains a primary source of income.

On a global level, it presents an innovative model for "agricultural fintech" and fractional asset ownership that can be applied to other regions and crops facing similar disconnects between small-scale farmers and dispersed capital. It contributes to knowledge sharing on using technology to enhance sustainability and efficiency in arid-land agriculture, a challenge relevant to many communities worldwide.

This document presents the initial proposal for Qinwan . We begin by detailing the significant real-world problem of inaccessible agricultural investment. Following this, we will elaborate on our proposed technological solution and its core vision. The document will then specify the product's

functional objectives for end-users, outline the project's technical and developmental goals, and define the key learning outcomes for the student team. The proposed development timeline is captured in a product roadmap, and the project's Scrum team is formally introduced. All references supporting this chapter are provided in the final section.

## 1.1 The Problem

In this section, we describe the challenges in agricultural investment, including the difficulties faced by investors and farm owners, and explain the need for a structured digital solution.

Many individuals are interested in agricultural investment but face difficulties entering this field because they do not own farmland or have enough capital to start or manage a farm. At the same time, many farm owners who own productive agricultural land struggle to find interested investors or individuals willing to invest in their farms. These farmers often have the land, expertise, and labor, but lack access to a reliable network of potential investors. As a result, they miss opportunities to expand their operations, improve their farming techniques, or increase their annual production. Agriculture is an important sector for the economy and food security[1][2], yet investment opportunities are often limited to large investors and farm owners, which reduces participation from the wider community and leaves many farmers with underutilized land.

A concrete example of this problem can be seen when a person wants to invest in date palm farming but does not own land and cannot afford to buy or operate a full farm. Even if they wish to invest with a farmer, they may not know which farms are available or how to invest safely. On the other side, a farm owner in Al-Ahsa or Al-Qassim may have 50,000 square meters of productive land with high-quality date palms, but no clear way to reach individuals who would be interested in leasing a portion of that land. As a result, many people lose interest in agricultural investment despite their willingness to participate, and many farmers continue operating below their full potential.

This project focuses on a specific part of this problem: the lack of an easy and organized way for individuals to invest in agriculture without needing land or large capital. It also addresses the absence of a clear digital platform that connects farm owners with investors and presents investment opportunities in a transparent way.

## 1.2 The Solution

In this section, we propose a solution to the problem described earlier. The proposed solution is a digital platform called Qinwan, which aims to facilitate fractional agricultural investment in date palm farms within Saudi Arabia.

Qinwan brings farm owners and investors together in one platform. Farm owners can register their farms and define specific land areas available for investment, each clearly displaying the total size, the types of date palms planted in that plot, and the expected seasonal yield. Investors can browse available farms through an interactive map of Saudi Arabia or a filterable list view. When selecting a farm, investors can view each available land plot and choose the exact area (in m<sup>2</sup>) they wish to lease based on the options provided by the farm owner.

Once an investor selects their desired land area, they can specify the investment duration and choose how they would like to benefit from the harvest at the end of the season: receiving the dates, earning a financial return, or donating the harvest to charity. The investment request is then submitted to the farm owner for review and confirmation.

The platform also allows users to track their investments through a personal dashboard that shows the land area they have leased, the types of date palms included in their plot, and the status of each active investment. In addition, Qinwan provides regular updates from the farms about the condition and progress of the palm trees, including photos and text updates sent directly by the farm owner. The system includes an interactive map that allows users to explore farms across different regions of Saudi Arabia.

This solution helps farm owners reach investors in a more organized and efficient way, while offering investors a transparent, flexible, and reliable agricultural investment experience with clear visibility into exactly which land and trees their money supports.

## 1.3 The Product

### 1.3.1 Product Vision

In this section, we define the product vision of Qinwan, clarifying what the platform aims to achieve for investors and farm owners and the value it intends to deliver.

For individual investors and date palm farm owners in Saudi Arabia,

Who seek a transparent, accessible, and impact-driven way to participate in the dates sector and directly support local agriculture,

Qinwan is a web-based agricultural investment platform

That enables users to discover, invest in, and monitor specific date palm trees on verified farms, transforming traditional agriculture into a transparent, trackable, and socially impactful digital asset class through a unified, accessible experience.

Unlike online marketplace platforms such as Haraj and Al-Aswaq Al-Mawsimiyah, which require prohibitive capital or operate through informal agreements that lack oversight and transparency,

Our product directly connects capital to specific assets on a map, provides verifiable farm and tree data, and ensures clear, flexible outcomes for every harvest building a trusted and modern investment ecosystem.[2]

## 1.3.2 Product Roadmap

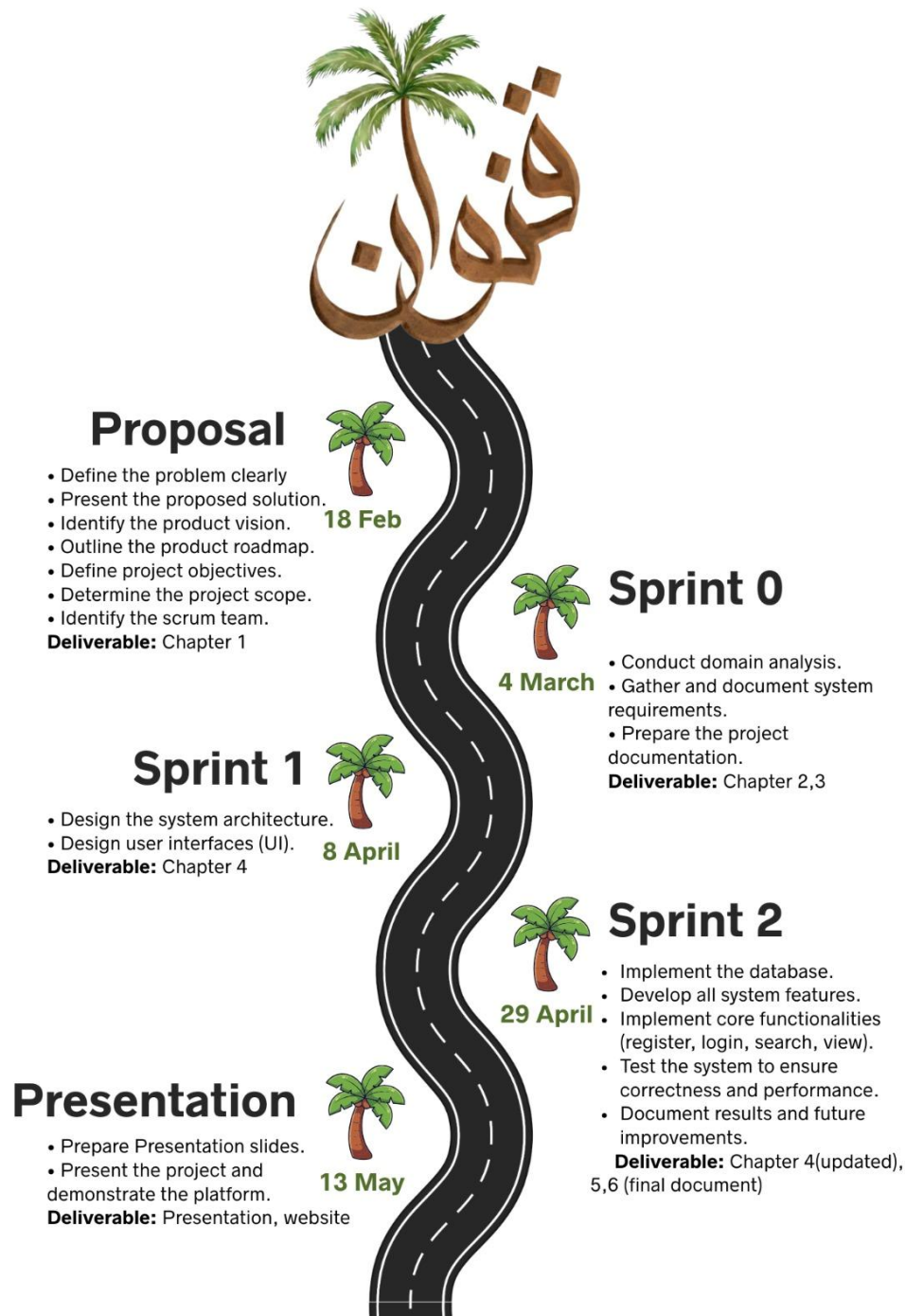


Figure 1 : Product roadmap



### 1.3.3 Objectives

In this section, we will define the product, project, and learning objectives pertaining to the development of the "Qinwan" web platform. These objectives represent the measurable goals and end-state outcomes we intend to achieve upon successful project completion.

#### **Project Objectives (Customer Focus - Value):**

##### **Farmers:**

#### **1. User Registration (Farmer)**

The user can create a new account by providing required information (name, email, phone number, password) and selecting the "Farmer" role to access farming features.

#### **2. User Login (Farmer)**

The user can log in using their registered email and password to access their farmer profile and dashboard.

#### **3. User Logout (Farmer)**

The user can securely log out of their account to end their session.

#### **4. Farm Submission for Approval**

The farmer can submit a detailed farm registration form including land size, location, palm tree types, and estimated annual yield for admin review and approval.

#### **5. Edit Farm Details**

The farmer can edit their farm information (e.g., description, photos, available land area) after approval.

#### **6. View Investment Requests**

The farmer can view a list of incoming investment requests from investors requesting to lease specific land areas within their farm.

#### **7. Accept Investment Requests**

The farmer can accept investor requests, confirming the lease agreement for the specified land area.

## **8. Reject Investment Requests**

The farmer can reject or cancel investment requests before or during the review process.

## **9. Send Periodic Farm Updates**

The farmer can send text and photo updates about crop conditions and palm trees to all investors who are currently leasing land on their farm.

## **10. View Total Leased Land Area**

The farmer can view the total land area (in m<sup>2</sup>) currently leased by investors on their dashboard.

## **11. View Number of Active Investors**

The farmer can view the total count of active investors currently leasing land on their farm.

## **12. View Total Expected Yield**

The farmer can view the estimated total yield for the current season based on leased land area.

## **13. View Estimated Profit Distribution**

The farmer can view projected profit calculations from current active investments.

### **For Individual Investors:**

#### **1. User Registration (Investor)**

The user can create a new account by providing required information (name, email, phone number, password) and selecting the "Investor" role to access investment features.

#### **2. User Login (Investor)**

The user can log in using their registered email and password to access their investor profile and dashboard.

#### **3. User Logout (Investor)**

The user can securely log out of their account to end their session.

#### **4. Browse Farms via Interactive Map**

The investor can discover farms accepting investment through an interactive map of Saudi Arabia.

#### **5. Browse Farms via List View**

The investor can discover farms accepting investment through a filterable list view.

#### **6. Filter Farms by Region**

The investor can filter available farms based on geographical region.

#### **7. Filter Farms by Farm Size**

The investor can filter available farms based on total farm area.

#### **8. Filter Farms by Date Type**

The investor can filter available farms based on the type of dates grown.

#### **9. Add Farm to Wishlist**

The investor can add farms to a favorites list for quick access later.

#### **10. View Farm Size**

The investor can view the total size of a selected farm.

#### **11. View Available Land Area for Investment**

The investor can view how much land area is currently open for investment on a specific farm.

#### **12. View Date Palm Types Grown**

The investor can view the types of date palms cultivated on the farm.

#### **13. View Number of Current Investors**

The investor can view how many investors are currently active on a specific farm.

#### **14. View User Ratings and Reviews**

The investor can view ratings and feedback left by other investors about the farm.

#### **15. Select Land Area for Investment**

The investor can select the farmer offer of a specific land area (in m<sup>2</sup>) within a farm to lease.

#### **16. Add Selected Land Area to Cart**

The investor can add their selected land investment item to the cart for review before submission.

#### **17. View Cart Items**

The investor can open their cart to view all selected land investment items.

#### **18. Remove Item from Cart**

The investor can delete or remove any investment item from their cart before submission.

#### **19. Edit Harvest Return Method in Cart**

The investor can edit the preferred harvest return method for any item while it is still in the cart.

#### **20. Select Home Delivery as Return Method**

The investor can choose to receive physical dates at their address after harvest.

#### **21. Select Sell Harvest as Return Method**

The investor can choose to have their share of the harvest sold and receive the financial profit.

#### **22. Select Donate Harvest as Return Method**

The investor can choose to donate their share of the harvest to charity.

#### **23. Specify Investment Duration in Cart**

The investor can set the lease duration (e.g., 6 months, 12 months) for each item while in the cart.

#### **24. Edit Investment Duration in Cart**

The investor can modify the lease duration for any item while it is still in the cart.

### **25. Submit Investment Request**

The investor submits all items from their cart as a single investment request.

### **26. View Pending Investment Requests**

The investor can view all investment requests currently waiting for farmer approval.

### **27. View Accepted Investment Requests**

The investor can view all investment requests that have been confirmed by the farmer.

### **28. View Cancelled Investment Requests**

The investor can view all investment requests that were rejected or cancelled.

### **29. Cancel Investment Request (Pre-Approval)**

The investor can cancel or delete their investment request only while its status is Pending.

### **30. View Leased Land Area in Dashboard**

The investor can view the total land area they have leased and its location within the farm.

### **31. View Farm Name and Farmer Contact**

The investor can view the farm name and farmer contact information in their dashboard.

### **32. View Date Palm Types in Leased Area**

The investor can view the types of date palms included in their leased land area.

### **33. View Expected Harvest Yield**

The investor can view the estimated harvest yield from their leased land area.

### **34. View Profit and Loss Summary**

The investor can view a summary of financial returns from their investments.

### **35. View Farmer Update Feed**

The investor can view a dedicated feed of text and photo updates received from the farmer about their leased land.

### **36. Request Edit Return Method (Post-Acceptance)**

The investor can request to change their chosen return method after the request is accepted, subject to farmer approval.

#### **For Administrators:**

##### **1. View All Registered Users**

The admin can view a complete list of all registered users on the platform.

##### **2. View All Registered Farms**

The admin can view a complete list of all farms registered on the platform.

##### **3. Review Farm Submissions**

The admin can review detailed farm registration forms submitted by farmers.

##### **4. Approve Farm Registrations**

The admin can approve farm submissions, making them visible to investors.

##### **5. Reject Farm Registrations**

The admin can reject farm submissions that do not meet platform requirements.

##### **6. Edit Farm Information**

The admin can edit or correct farm details when necessary.

##### **7. Remove Farm Listings**

The admin can remove farms from the platform if violations occur.

##### **8. View All Investment Transactions**

The admin can view a complete record of all investment transactions across the platform.

##### **9. View User Activity Logs**

The admin can view activity logs for both farmers and investors.

##### **10. Edit User Content**

The admin can edit or remove inappropriate content including farm descriptions and user reviews.

### **11. Receive User Complaints**

The admin can receive and review complaints reported by users.

### **12. View Total Registered Users Count**

The admin can view the total number of users registered on the platform.

### **13. View Total Active Farms Count**

The admin can view the total number of approved and active farms.

### **14. View Total Invested Land Area**

The admin can view the total land area currently under investment across all farms.

### **15. View Total Transaction Volume**

The admin can view the total value of all investment transactions processed.

## **Project Objectives (solution focus–plan):**

These objectives describe the stages required to complete the Qinwan platform:

1. Conduct interviews and questionnaires with potential users (investors and farmers).
2. Collect and analyze data related to date palm farming and agricultural investment (domain analysis).
3. Identify user requirements and apply requirements elicitation techniques.
4. Design user interfaces for the Qinwan platform.
5. Design a database for users, farms, palm trees, and investments.
6. Develop and implement the platform and its core features.
7. Integrate interactive map features to display farm locations.
8. Test the platform to ensure functionality and usability.
9. Deploy the platform and prepare project documentation.

### Learning Objectives (student focus):

The following objectives outline what the team will learn from completing this project:

- Learn a new programming language, e.g., XML.
- Use new tools, such as GitHub and Jira.
- Learn and implement the Agile development methodology through sprint-based project work.
- Learn how web applications connect to databases to store and retrieve user information.
- Practice software documentation throughout the project lifecycle.

By achieving these objectives, our Qinwan platform will contribute to supporting digital transformation in the agricultural investment sector while helping our team develop practical skills in web-based application development.

In the next section, we will discuss the scope of Qinwan and explore the specific functionalities and features it will provide to users.

#### 1.3.4 Scope

In this section, we define the scope of Qinwan by clarifying what features and functionalities are included in the project and what falls outside its boundaries.

Qinwan is a web-based digital agricultural investment platform designed for individual investors and date palm farm owners in Saudi Arabia aged 18 years and older, with basic to average technical skills. The platform is built as a fully responsive web application, adapting seamlessly to different screen sizes and devices, delivering an optimized experience on desktop, tablet, and mobile through modern web browsers (Chrome, Safari, Edge). The user interface and all platform content are in Arabic only. The system supports three defined user roles: Investor, Farmer, and Administrator, each with distinct access rights, and does not support guest browsing without registration. Core functionality includes account creation, farm browsing and filtering, fractional land-area investment, investment tracking via a personal dashboard, harvest yield management (receive, sell, or donate), and farm update feeds. The platform does not include real-time IoT farm monitoring, external payment gateway integration, advanced financial analytics, legal contract enforcement, government system integration, or automated smart contract execution. Any features beyond those defined in the objectives are considered future enhancements and fall outside the scope of this project.



## 1.4 The Scrum Team

| Scrum Team           |                                                                 |
|----------------------|-----------------------------------------------------------------|
| <b>Product Owner</b> | Noora Aluqaili.                                                 |
| <b>Developers</b>    | Noora Aluqaili , Jana Alothman , Haya Alhajri , Noura Altuwiam. |
| <b>Scrum Master</b>  | I. Ghaida ALFayez.                                              |
| <b>Stakeholders</b>  | I. Ghaida ALFayez.                                              |

Table 1 : Scrum Team

## 2 Chapter 2: Domain Analysis

### 2.1 Literature Review

#### Competitor Haraj [3]:

Brief Description: Haraj is a popular Saudi online marketplace that allows users to buy and sell various products, including land and farms. Some users post agricultural lands or farms for sale or lease.

Strengths:

- 1- Large user base across Saudi Arabia.
- 2- Easy posting and browsing of listings.
- 3- Covers many categories including farms.

Limitations:

- 1- Not specialized in agricultural investment.
- 2- No fractional investment option (users must buy full property).
- 4- No structured system connecting investors with farm owners.

### Competitor Atyab Al-Tamor[4] :

Brief Description : Atyab Al-Tamor is an online store specialized in selling dates and related products in Saudi Arabia. The platform provides various types of packaged dates for individual customers and gift orders.

Strengths:

1. Specialized in date products.
2. Professional product presentation and packaging.
3. Online ordering and delivery services.

Limitations:

1. Focuses only on selling final date products.
2. Does not provide palm tree investment opportunities.
3. No transparency regarding farm-level ownership tracking.

| Features                        | Haraj | Atyab Al-Tamor | Qinwan |
|---------------------------------|-------|----------------|--------|
| Palm tree investment            | X     | X              | ✓      |
| Agricultural land reservation   | ✓     | X              | ✓      |
| Investment tracking system      | X     | X              | ✓      |
| Structured & transparent system | X     | X              | ✓      |

Table 2 : Competing Platforms and Their Features Compared with Qinwan

Based on the comparison table above, it is clear that existing platforms such as Haraj and Atyab Al-Tamor mainly focus on agricultural listings or product sales, without providing a structured investment framework. While some platforms may support land reservation, they do not offer palm tree-level investment or a dedicated tracking system.

In contrast, Qinwan integrates palm tree investment, agricultural land reservation, and a built-in investment tracking system within a structured digital environment. The platform uniquely connects investors to specific palm trees and enables organized monitoring of their investment lifecycle. This combination of transparency, structure, and asset-level visibility clearly differentiates Qinwan from existing competitors and positions it as a comprehensive agricultural investment solution.

## 3 Chapter 3: Requirements Engineering

2

3

### 3.1 Requirements Elicitation and Analysis

To identify the system requirements for Qinwan, several requirement elicitation methods were used. First, online questionnaires were distributed to two target groups: potential investors and date palm farm owners. The questionnaires aimed to understand user expectations, preferred features, and challenges in agricultural investment.

In addition, informal interviews were conducted with farm owners to better understand their needs regarding land leasing, investment management, and transparency requirements. Market research was also performed by analyzing similar platforms such as Haraj and Atyab Al-Tamor to identify existing gaps and limitations.

Furthermore, official reports related to date production and agricultural development in Saudi Arabia were reviewed to ensure that the proposed system aligns with market needs and local agricultural practices.

The collected data was analyzed and translated into functional and non-functional requirements that define the core features of the Qinwan platform.

### 3.1.1 Interviews

#### Participants' information

| Name        | Age      | Nationality | gender |
|-------------|----------|-------------|--------|
| 1. Mohammed | 55 Years | Saudi       | Male   |
| 2. Ibrahim  | 50 Years | Saudi       | Male   |
| 3. Abdullah | 47 Years | Saudi       | Male   |
| 4. Saad     | 60 Yeas  | Saudi       | Male   |

Table 3 : Interview participants' information

#### Interviews results and findings

Interviews were conducted with two investors and two farmers to understand their needs and expectations for a fractional agricultural investment platform.

From the investors' side, experience level affected their expectations. As shown in *Appendix A (Investor 1, Q1–Q3)*, Investor 1 has previous experience in agriculture and knows the importance of regular follow-up and paying attention to environmental factors such as water quality and soil fertility. In contrast, Investor 2, as indicated in *Appendix A (Investor 2, Q1–Q2)*, has no previous experience but is interested in investing without owning land. This shows that the platform can attract both experienced and new investors.

Both investors said that clear information is very important before making any investment decision. As shown in *Appendix A (Investor 1, Q3)*, Investor 1 focused more on the farm's environmental conditions, while *Appendix A (Investor 2, Q3)* indicates that Investor 2 focused on profit percentage and operational details. Both wanted the platform to be simple and easy to use, as reflected in *Appendix A (Investor 1, Q4; Investor 2, Q4)*. They also highlighted the importance of direct communication with farmers and regular visual updates to help build trust, as shown in *Appendix A (Investor 1, Q5; Investor 2, Q5)*.

From the farmers' side, both participants showed strong agricultural experience, as shown in *Appendix A (Farmer 1, Q1; Farmer 2, Q1)*, and were open to external investment to reduce financial pressure and increase production, as indicated in *Appendix A (Farmer 1, Q2; Farmer 2, Q2)*. They are willing to share detailed information about their farms, such as expected production, water sources, infrastructure, and operating costs, as shown in *Appendix A (Farmer 1, Q3; Farmer 2, Q3)*. However, they also emphasized that the platform should be easy to use, as reflected in *Appendix A*

(Farmer 1, Q4; Farmer 2, Q4). One farmer mentioned that high commission fees might make him stop using the platform, as shown in *Appendix A (Farmer 2, Q4)*, which shows that fair pricing is important.

Overall, the interviews show that investors and farmers have similar expectations. The success of the platform depends on clear information, easy use, fair pricing, and regular communication that builds trust between both sides.

### 3.1.2 Questionnaire

#### Executive Summary

A total of 81 participants completed two separate questionnaires for the Qinwan platform between February 27–28, 2026. The investor survey (n=45) and farmer survey (n=36) together reveal a consistent and actionable picture of user needs.

The majority of investors (62%) are aged 46 and above, indicating a mature, financially capable audience who expect a simple and trustworthy experience. When it comes to browsing farms, 64% want both a map view and a list view available simultaneously. Farm photos and short video updates are the most demanded content types, with photo importance scoring an average of 4.3 out of 5. On the farmer side, 69% have never leased their land before but are actively interested, meaning Qinwan is entering a largely untapped market. Both user groups independently chose WhatsApp as their preferred notification channel (73% of investors, 75% of farmers), and both want visual content in farm updates (photos and short videos). The single most repeated insight across both surveys: users want transparency, verification, and trust between investors and farm owners. Features such as investor ratings, farm activity feeds, and a clear dashboard are not nice to have they are core expectations.

## A. Part 1 —Investor Survey (n = 45)

### Key Findings at a Glance

Map + List dual view, palm type and farm rating filters, photo/video upload for updates, WhatsApp notifications, harvest sale & profit transfer system, investor rating system, and automated response reminders for farmers.

| Question                       | Key Finding                                |
|--------------------------------|--------------------------------------------|
| Age group                      | 46 and above — 62%                         |
| Preferred farm browsing        | Both Map + List — 64%                      |
| Top filter criteria            | Farm rating / Palm type — 29 participants  |
| Photo importance (Likert)      | Average 4.3 / 5 — 60% gave full score of 5 |
| Preferred update type          | Short video clips — 71%                    |
| Preferred notification channel | WhatsApp — 73%                             |
| Preferred harvest method       | Sell harvest & receive profits — 67%       |

Table 4 : Summary of Key Findings from the Investor Survey

### Q1 — What is your age group?

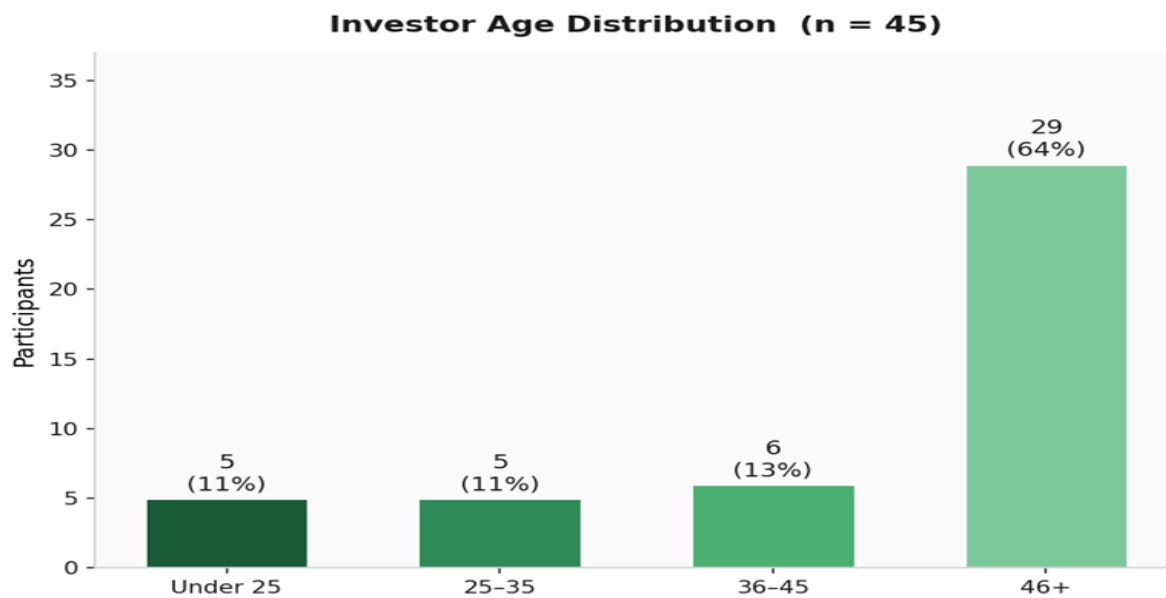


Figure 2 : Questionnaire Results (Q1 | Investor)

The dominant age group is 46+ (62%), followed by 36–45 (16%). This means the platform's primary users are experienced, financially established individuals. The UI must be clean and straightforward, avoiding overly technical language or complex navigation patterns.

## Q2 — How do you prefer to browse farms?

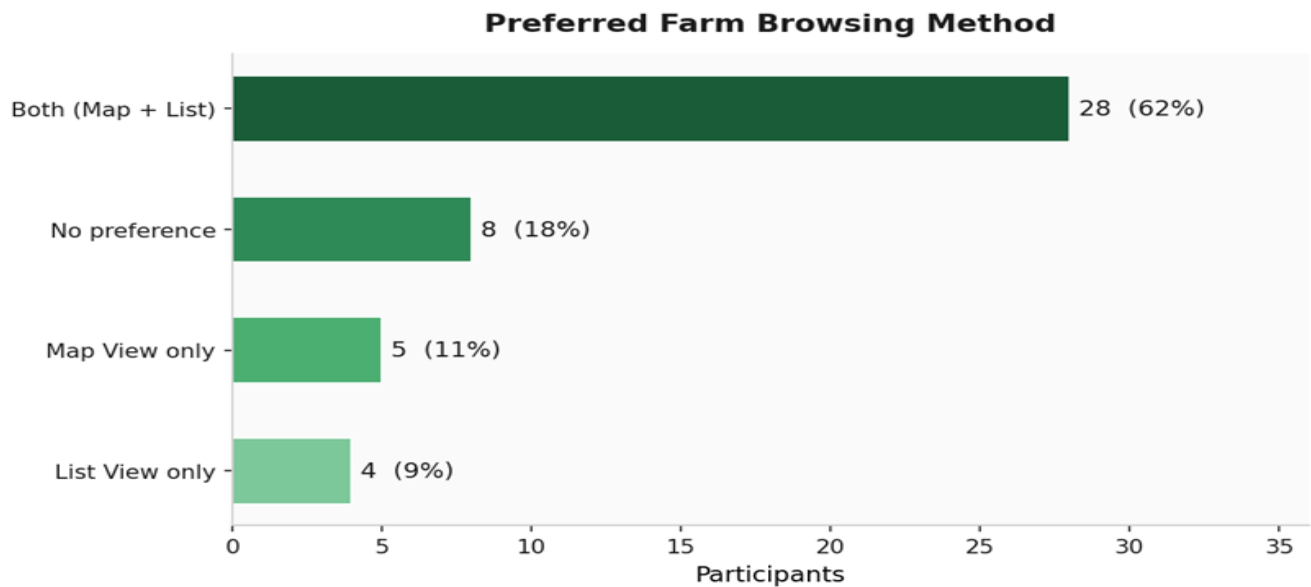


Figure 3 : Questionnaire Results (Q2 / Investor)

64% want both views available (Map + List), and only 9% prefer Map alone or List alone. Offering just one view would alienate most users. Both views must be built and accessible from the same screen.

## Q3 — What are your most important filter criteria? (multiple choice)

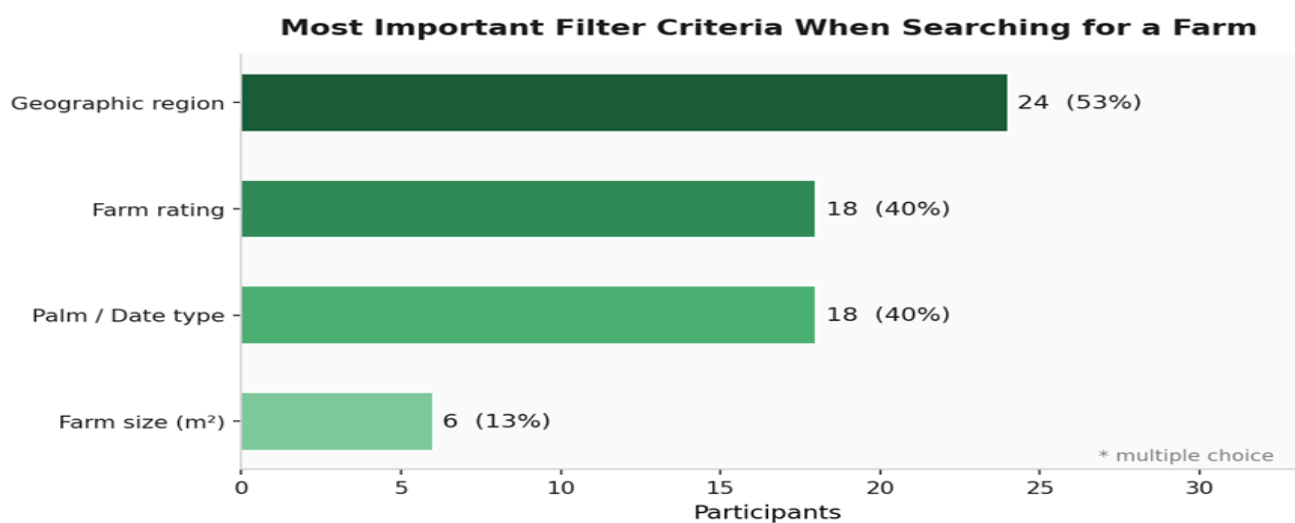


Figure 4 : Questionnaire Results (Q3 / Investor)

Geographic region (53%) leads, closely followed by palm/date type and farm rating (both 40%). Farm size scored notably lower at 13%, suggesting investors care more about what is grown and where, rather than raw acreage. The filter by palm type must be a prominent, first-class feature in the UI.

#### Q4 — How important are high-quality farm photos before investing? (1–5 scale)

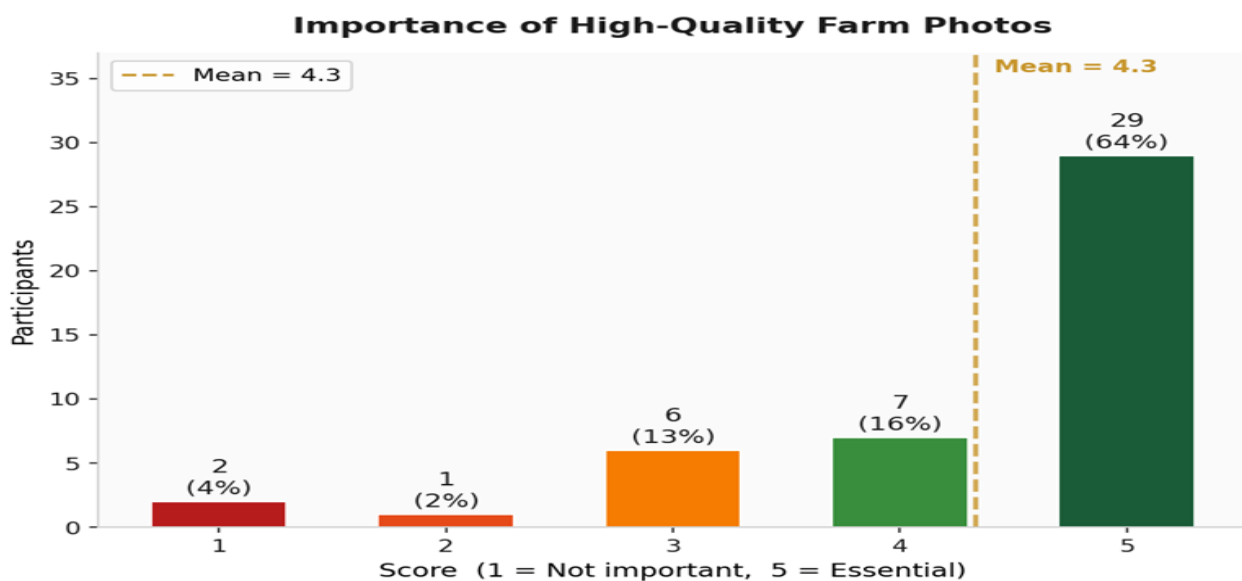


Figure 5 : Questionnaire Results (Q4 / Investor)

Average score: 4.3/5. A striking 60% gave the maximum score of 5, and only 6% rated this below 3. Photos are not cosmetic .they are a critical trust signal that directly influences investment decisions. Farmers must be required (or strongly encouraged) to upload quality photos when registering their farm.



### Q5 — What type of farm updates do you want to receive? (multiple choice)

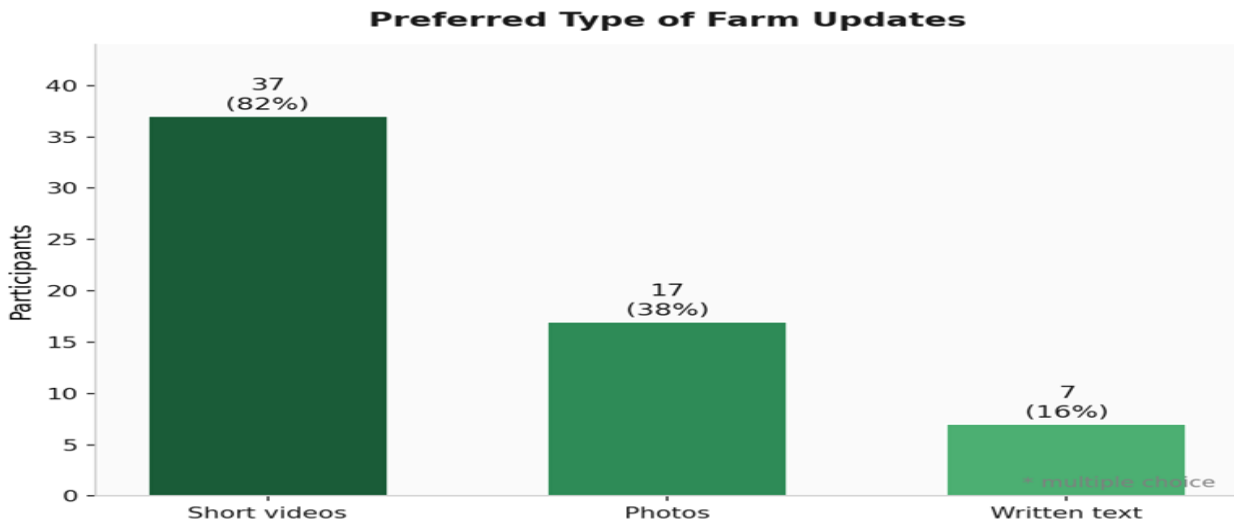


Figure 6 : Questionnaire Results (Q5 / Investor)

Short video clips dominate at 71%, followed by photos (38%) and text (27%). This directly aligns with farmer preferences in Part 2. The update feature must support video uploads, not just text or images. Both sides of the platform independently arrived at the same conclusion.

### Q6 — How do you prefer to receive platform notifications?

#### Preferred Notification Channel — Investor

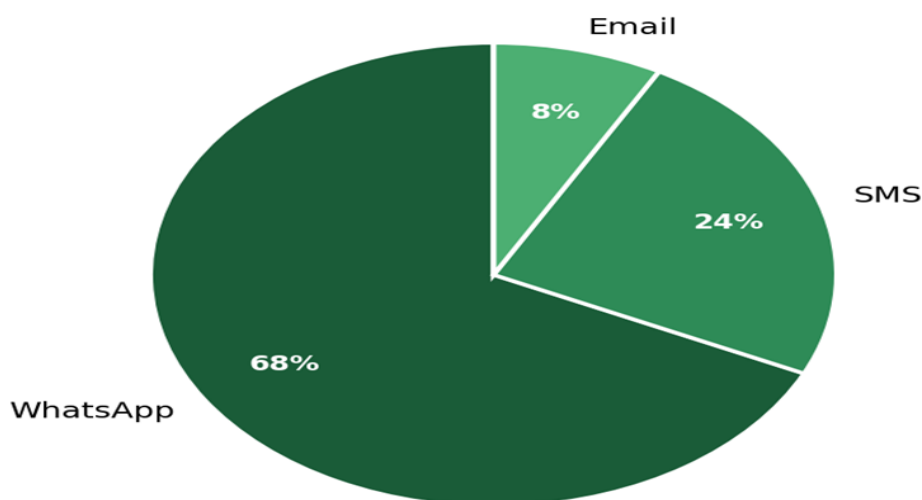


Figure 7 : Questionnaire Results (Q6 / Investor)

WhatsApp accounts for 73% of preferences. SMS and email are distant second choices. Integrating with WhatsApp Business API should be treated as a high-priority infrastructure requirement, not an optional feature.

### Q7 — How do you prefer to benefit from the harvest?

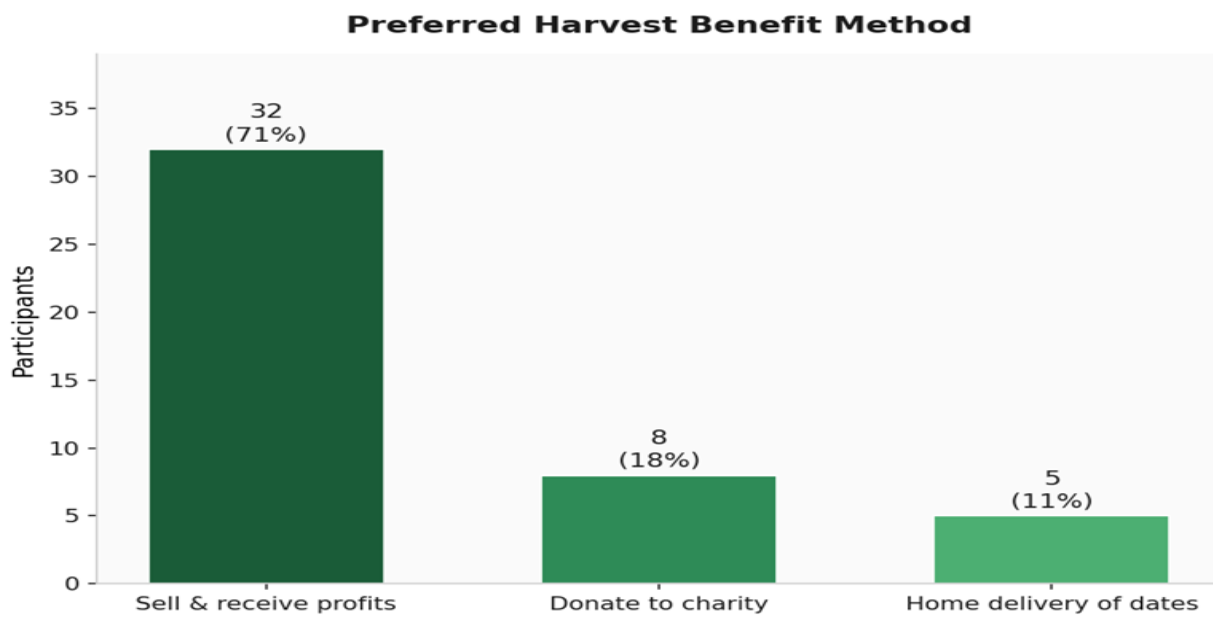


Figure 8 : Questionnaire Results (Q7 / Investor)

67% want to sell the harvest and receive cash profits . the classic return on investment model. Only 18% want physical date delivery, and 13% prefer to donate. This confirms the platform must build a robust harvest sale and profit transfer system as a primary use case.

### Open-Ended Question — Investor Suggestions

| # | Response                                                                                                                                                                     |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Would love a simple investment dashboard showing farm status and production, with photo/video reports for transparency, plus a direct messaging feature inside the platform. |
| 2 | Suggestion: add 'water availability' as a farm filter criterion.                                                                                                             |
| 3 | Want to see estimated profit and loss breakdowns before committing to an investment.                                                                                         |
| 4 | Focus on ensuring qualified labor on farms.                                                                                                                                  |
| 5 | Option to split the harvest between selling, donating, and home delivery would be valuable.                                                                                  |

|   |                                                                                                   |
|---|---------------------------------------------------------------------------------------------------|
| 6 | Farm detail pages should include road type (paved/dirt), groundwater availability, and soil type. |
| 7 | Would like to see harvest statistics and expected crop count estimates.                           |
| 8 | Groundwater depth and abundance should be a filter/detail field for farms.                        |

Table 5 : Investor Suggestions

The most repeated theme in open responses: environmental farm data (water, soil, road access) is a missing but desired filter. This is not currently planned and represents a potential future feature or an enhancement to the farm registration form.

## B. Part 2 —Farmer Survey (n = 36)

### Key Findings at a Glance

| Question                       | Key Finding                                               |
|--------------------------------|-----------------------------------------------------------|
| Prior leasing experience       | <b>69% interested but never leased before</b>             |
| Most wanted investor info      | <b>Full name &amp; ID + Area &amp; Duration — highest</b> |
| Preferred response window      | <b>One full week — 44%</b>                                |
| Update content type            | <b>Photos — 69%</b>                                       |
| Update frequency               | <b>Monthly — 33%</b>                                      |
| Preferred notification channel | <b>WhatsApp — 75%</b>                                     |
| Most wanted dashboard info     | <b>Estimated profits + investor name list — highest</b>   |

Table 6 : Summary of Key Findings from the Farmer Survey

### Q1 — Have you previously leased part of your farm?

#### Have You Previously Leased Part of Your Farm?

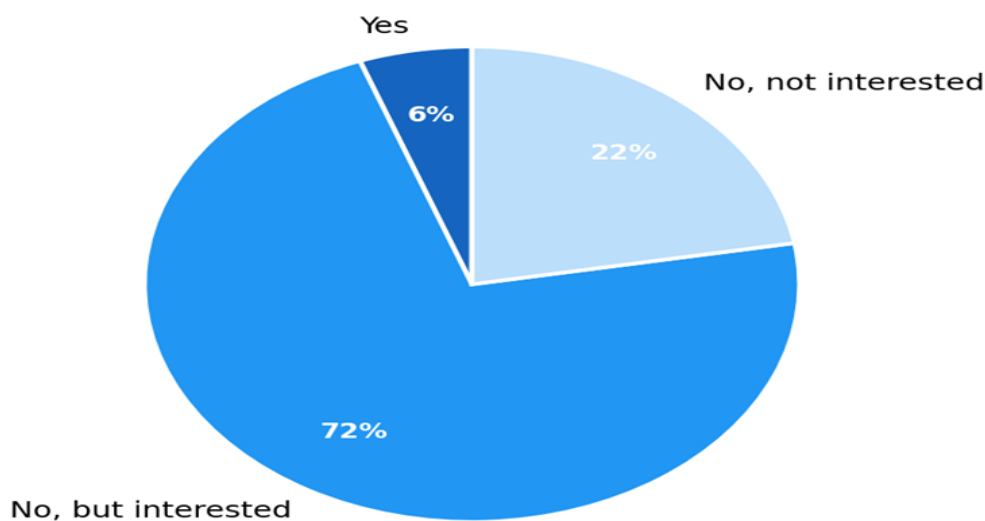


Figure 9 : Questionnaire Results (Q1 / Farmer)

69% are interested but have never leased before ,Qinwan is entering an untapped market. These first-time users will need an onboarding flow that clearly explains the leasing process and builds confidence in the platform from day one.

## Q2 — What investor information do you need when receiving a request? (multiple choice)

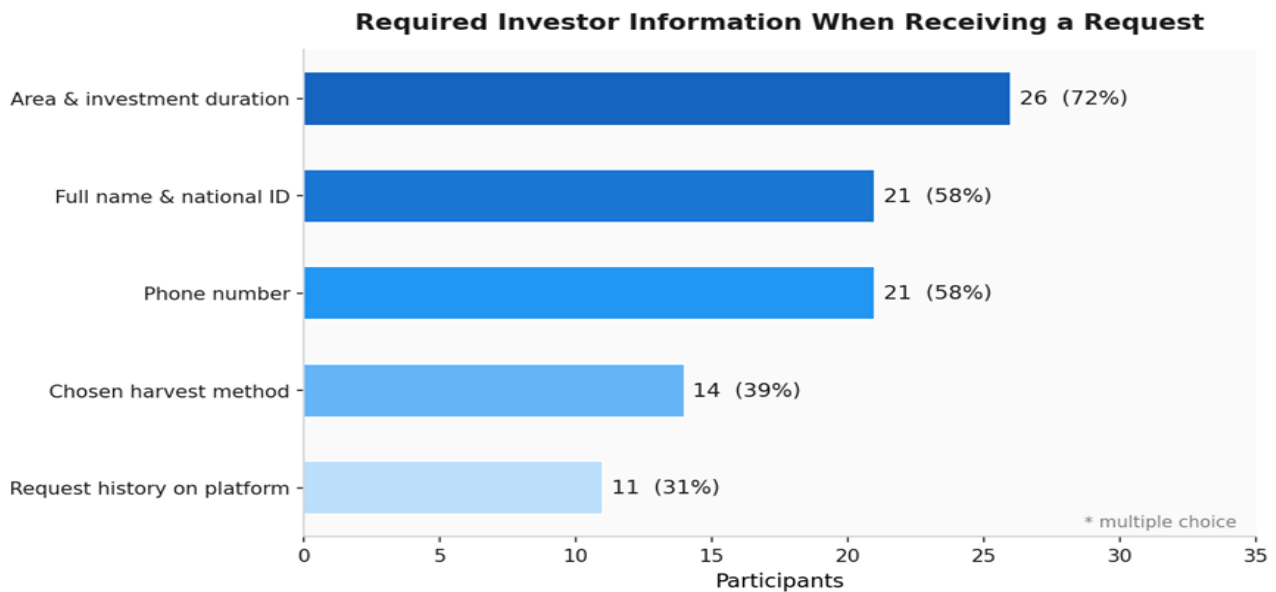


Figure 10 : Questionnaire Results (Q2 / Farmer)

Full area & duration (72%) leads, followed by full name & ID and phone number (both 58%). Farmers want identity verification before they approve any investment. The investor profile page must surface this information clearly and upfront.

## Q3 — How long do you need to respond to an investment request?

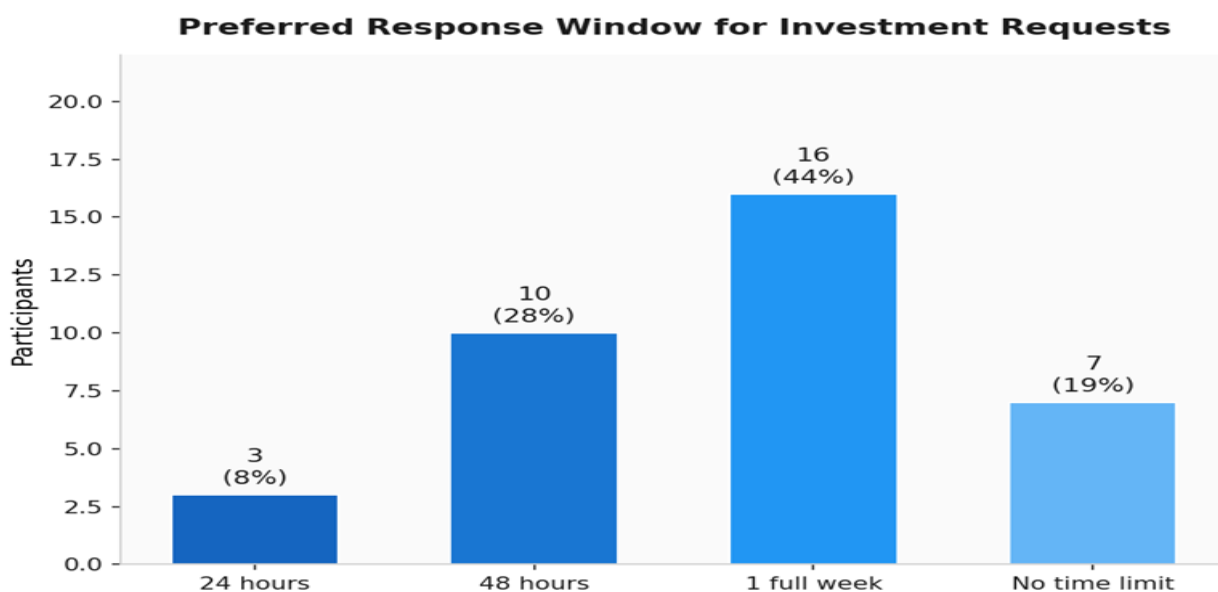


Figure 11 : Questionnaire Results (Q3 / Farmer)

44% need a full week to respond. The system must include automated reminders sent to the farmer as the deadline approaches , otherwise requests will expire silently and investors will be left waiting with no feedback.

#### Q4 — What content will you share in periodic updates? (multiple choice)

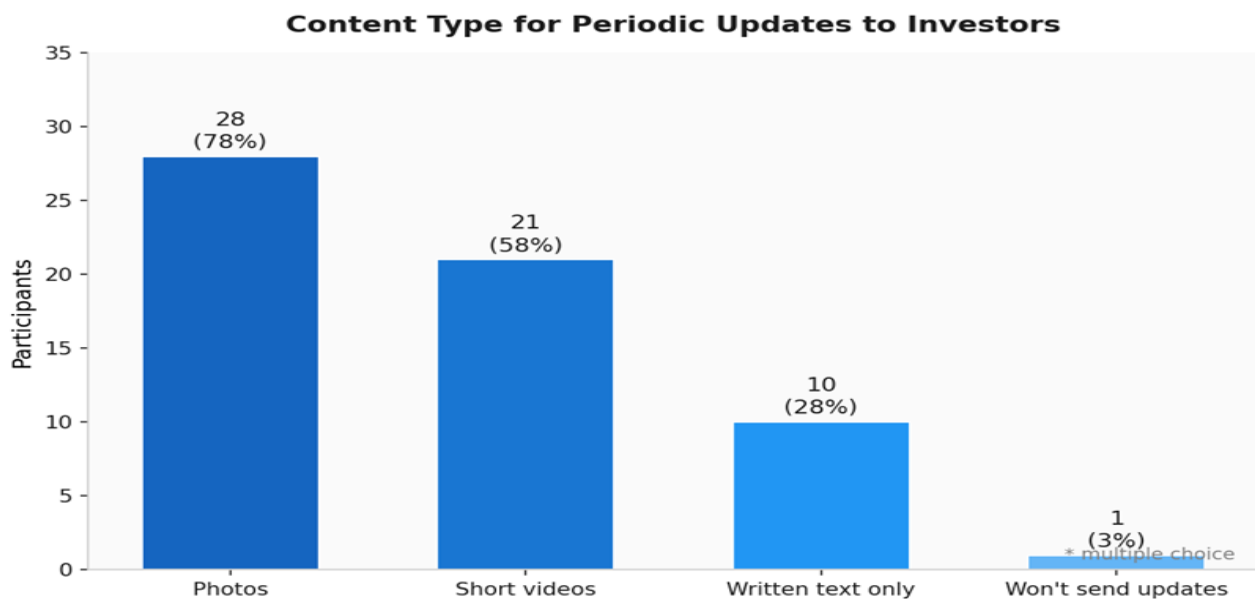


Figure 12 : Questionnaire Results (Q4 / Farmer)

Photos lead at 69%, followed by video (58%) and text+photos (42%). This is consistent with investor preferences , both sides want visual content. The update posting feature must make it easy to attach photos and videos directly from a mobile device.

### Q5 — How often will you send updates to investors?

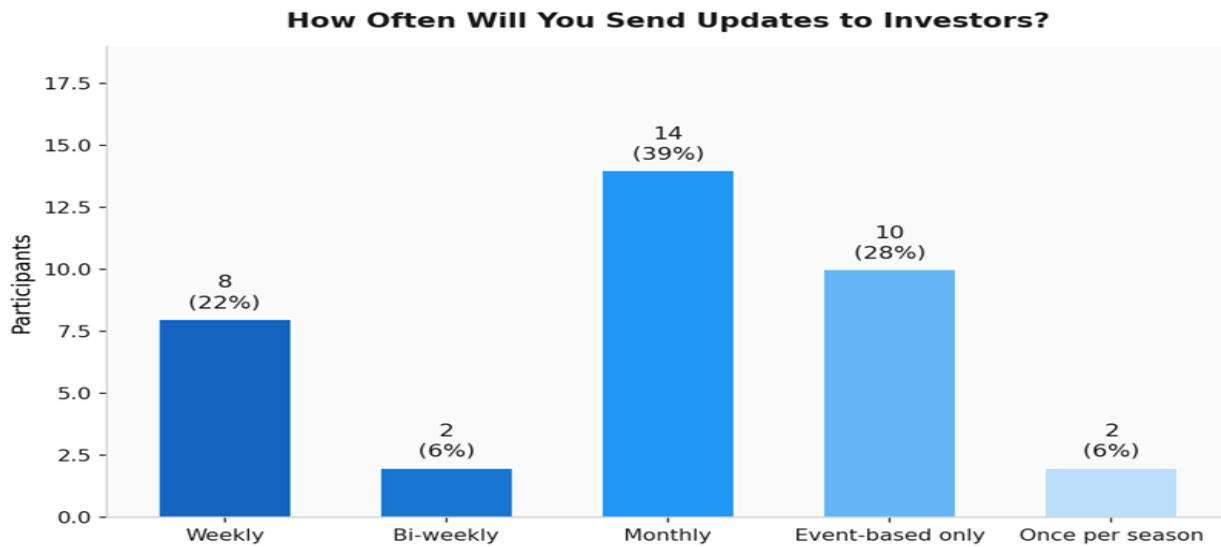


Figure 13 : Questionnaire Results (Q5 / Farmer)

Monthly (33%) and event-driven updates (25%) are the top two answers combined account for 58% of farmers. The platform should allow farmers to schedule recurring updates and trigger event-based ones (e.g., 'harvest started'). A gentle monthly reminder to post an update would improve investor experience significantly.

### Q6 — How do you prefer to receive platform notifications?

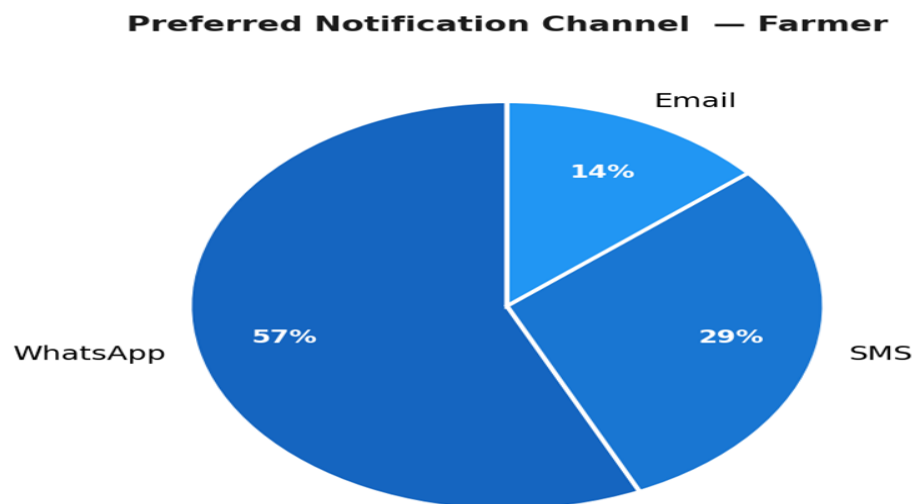


Figure 14 : Questionnaire Results (Q6 / Farmer)

WhatsApp is the top choice at 75% , matching investor preference at 73%. This cross-survey alignment makes WhatsApp integration the single highest-priority notification infrastructure decision for the entire platform.

### Q7 — What information do you want in your farm dashboard? (multiple choice)

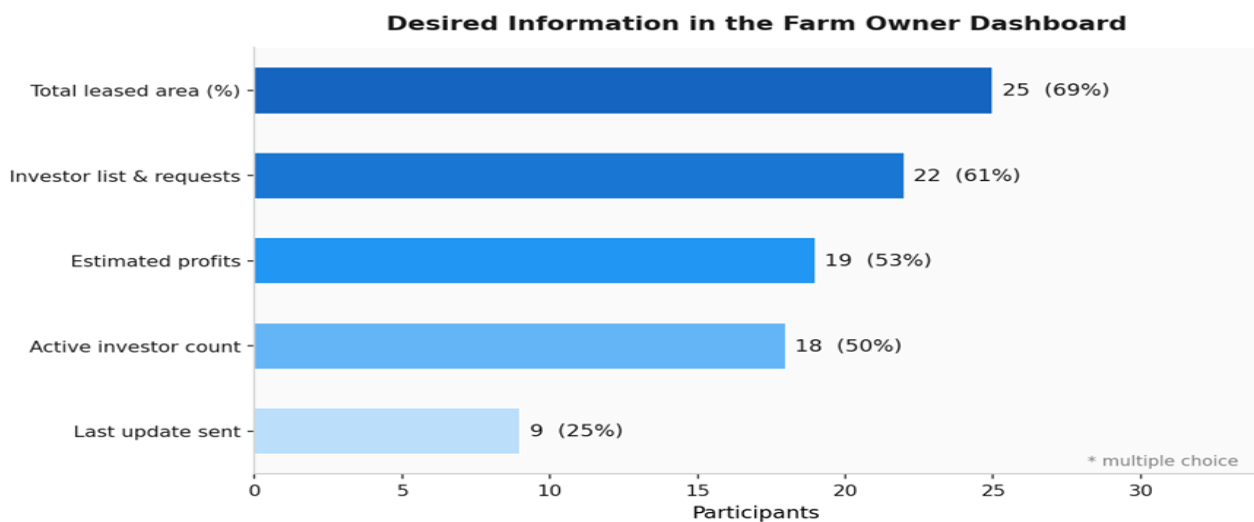


Figure 15 : Questionnaire Results (Q7 / Farmer)

Farmers want a financial + operational overview: estimated profits, list of investors and their requests, and the number of active investors are the top choices. The dashboard must act as both a management tool and a business summary, not just a status page.

### Open-Ended Question — Farmer Challenges

| # | Response                                                                                                                                                                                                             |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Hardest challenge: finding trustworthy investors who follow through. Qinwan can help by providing a vetted investor base, a smart tracking dashboard, and periodic updates — increasing transparency for both sides. |
| 2 | Finding serious tenants who pay on time and treat the farm with care is the main challenge.                                                                                                                          |
| 3 | Farm ownership documentation (shared deeds, old paper titles) may complicate registration on the platform.                                                                                                           |
| 4 | There is no dedicated platform for farm leasing in Saudi Arabia — Qinwan fills this gap directly.                                                                                                                    |
| 5 | Marketing the farm and finding investors is the main challenge. Confident that Qinwan can solve this.                                                                                                                |



|   |                                                                                                                                                                  |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | Collecting rent from tenants is a recurring obstacle — the platform's payment system should address this.                                                        |
| 7 | Suggest expanding investment types: buying dates from palm trees, full farm leasing, seasonal harvesting, etc.                                                   |
| 8 | Finding an investor who pays, respects the farm, and returns it in the same condition is the hardest part — an investor rating system would directly solve this. |

Table 7 : Farmer Challenges

The most frequently mentioned challenge across farmer responses: finding a trustworthy, serious investor. An investor rating and review system, combined with a verified identity requirement, would directly address this and increase farmer confidence in using the platform.

### Summary of Requirements Elicitation Findings

The system requirements were identified using surveys and interviews with potential investors and farm owners. The survey results highlighted key user needs such as browsing farms using map and list views, receiving visual updates through photos and videos, and using WhatsApp for notifications.

The interviews provided deeper insights, showing that investors require clear farm information such as water availability, soil quality, and expected profit, while farm owners emphasized the importance of simple platform usage and sharing farm updates with investors.

### Confirmed Features by Priority

| Priority | Feature                                                | Evidence                                                                                                                                          |
|----------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| High     | <b>Map View + List View (dual browsing)</b>            | 64% of investors — both views required                                                                                                            |
| High     | <b>Filter by palm/date type &amp; farm rating</b>      | Top two filter criteria across investors                                                                                                          |
| High     | <b>High-quality photo &amp; video upload for farms</b> | 4.3/5 avg importance + 71% want video updates; interviews confirmed preference for visual farm monitoring through photos, videos, or farm visits. |
| High     | <b>WhatsApp notifications</b>                          | 73% investors, 75% farmers prefer WhatsApp; interviews highlighted the need for direct communication between investors and farmers.               |

|        |                                                         |                                                                                                                                                                                           |
|--------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| High   | <b>Harvest sale &amp; profit transfer system</b>        | 67% of investors prefer sell & receive profits                                                                                                                                            |
| Medium | <b>Investor rating &amp; review system</b>              | Repeated in 4+ open-ended farmer responses                                                                                                                                                |
| Medium | <b>Automated farmer reminder before request expires</b> | 44% need a full week — silent expiry is a risk                                                                                                                                            |
| Medium | <b>Environmental farm filters (water, soil, road)</b>   | Mentioned by 4+ investors in open responses; interviews confirmed the importance of environmental information such as water availability, soil fertility, and palm type before investing. |
| Low    | <b>Harvest split option (sell + donate combined)</b>    | Suggested by 1 investor as a nice-to-have                                                                                                                                                 |

Table 8 : Confirmed Features by Priority

## 3.2 System Users

This section describes the detailed characteristics of the main users of the platform. The system includes two main types of users: individual investors and farm owners who participate in agricultural investment through the platform.

### 1. Individual Investors

- Mainly **Saudi citizens or residents** interested in local agricultural investment.
- May have **different educational levels**.
- May **have previous experience in agriculture or may be new to the sector**, as some investors are interested in passive investment without managing the farm directly.
- Have **basic to intermediate technical knowledge** in using websites and mobile applications.
- Usually use **smartphones or personal computers** to access the platform.
- Usually interested **in agricultural investment opportunities** and following farm performance through the platform

### 2. Farm Owners

- Typically adults aged between **30 and 65 years**.
- Mainly **Saudi farm owners** who manage palm farms.
- May have **different educational levels**.
- Have **strong experience in agriculture and palm farming**, including farm management and production knowledge.
- Have **basic to intermediate technical knowledge** in using websites and mobile applications.
- Usually use **smartphones** to access the platform.
- Aim to **promote their farms, attract reliable investors, increase production, and generate additional revenue** through the platform.

### 3.3 Use Case Diagram

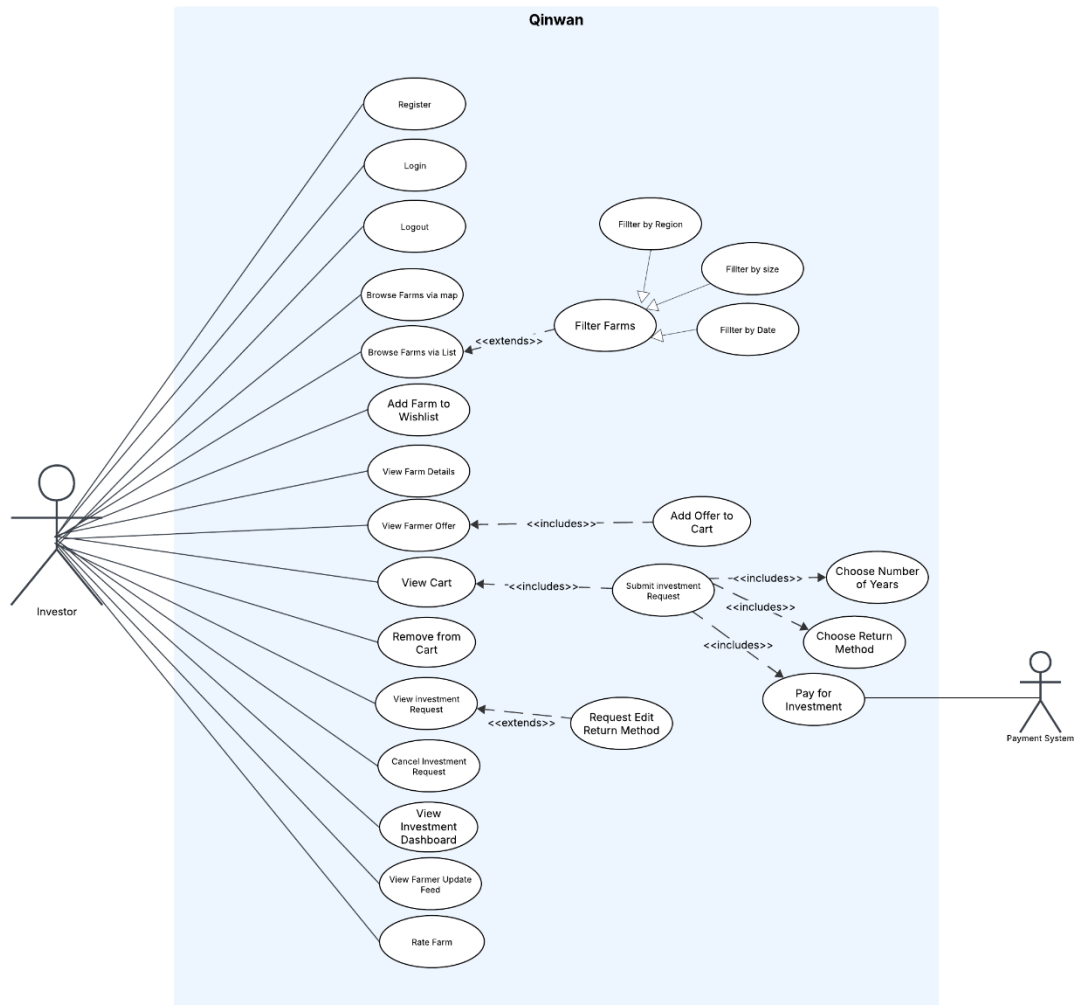


Figure 16 : Use Case Diagram -1

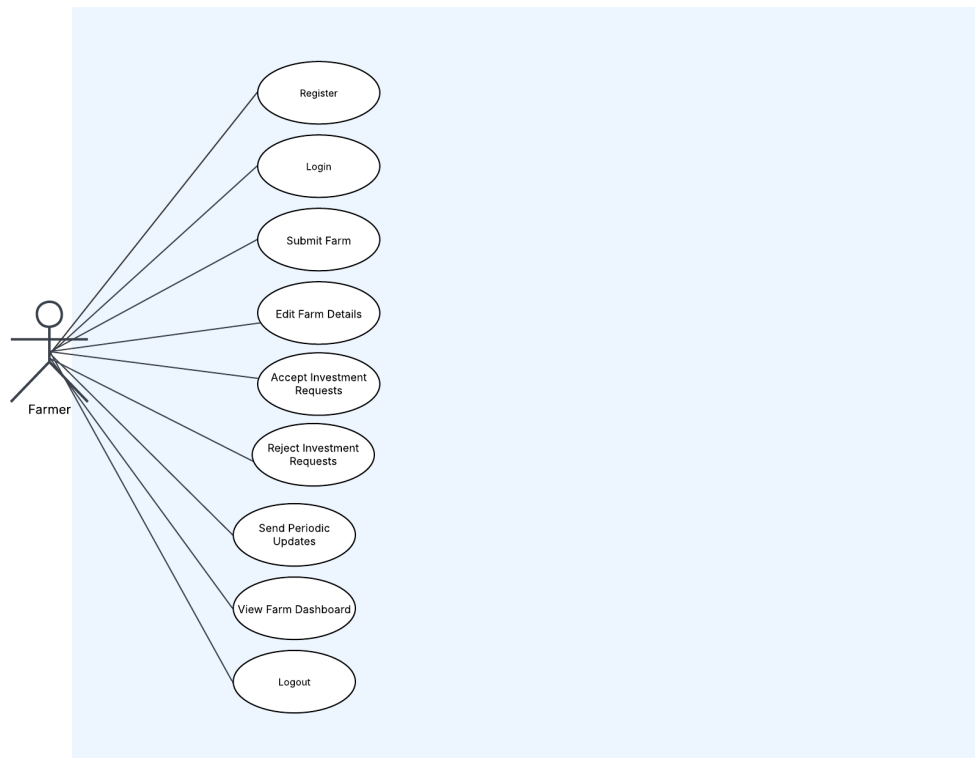


Figure 17 : Use Case Diagram -2

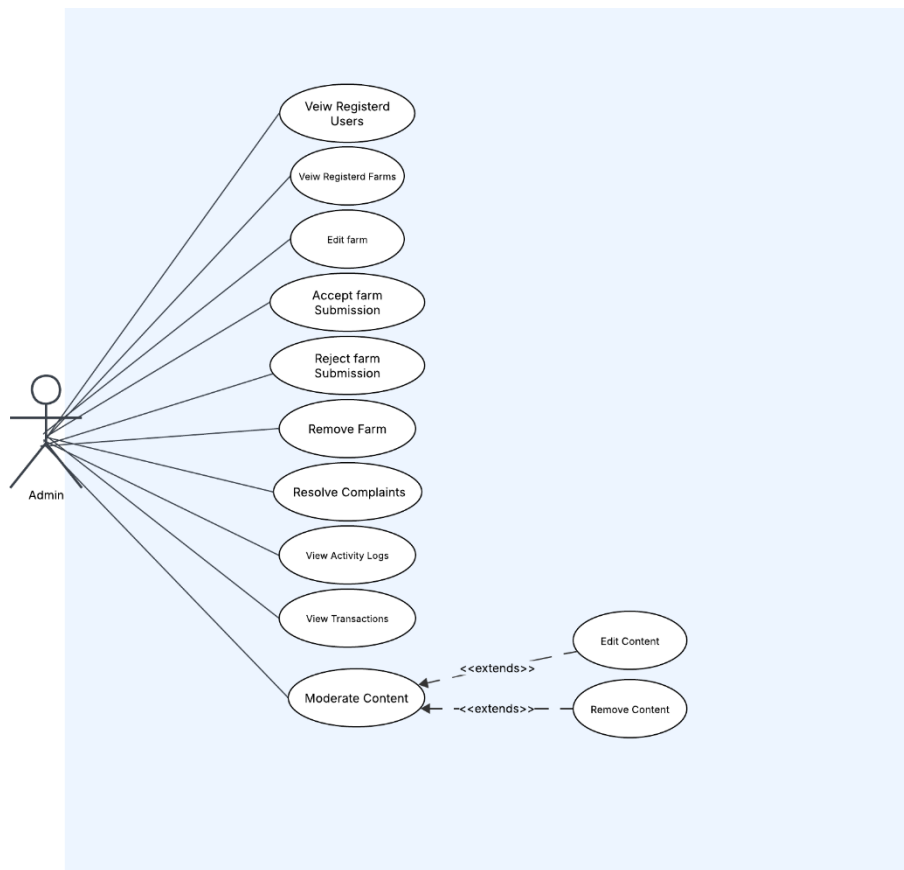


Figure 18 : Use Case Diagram -3

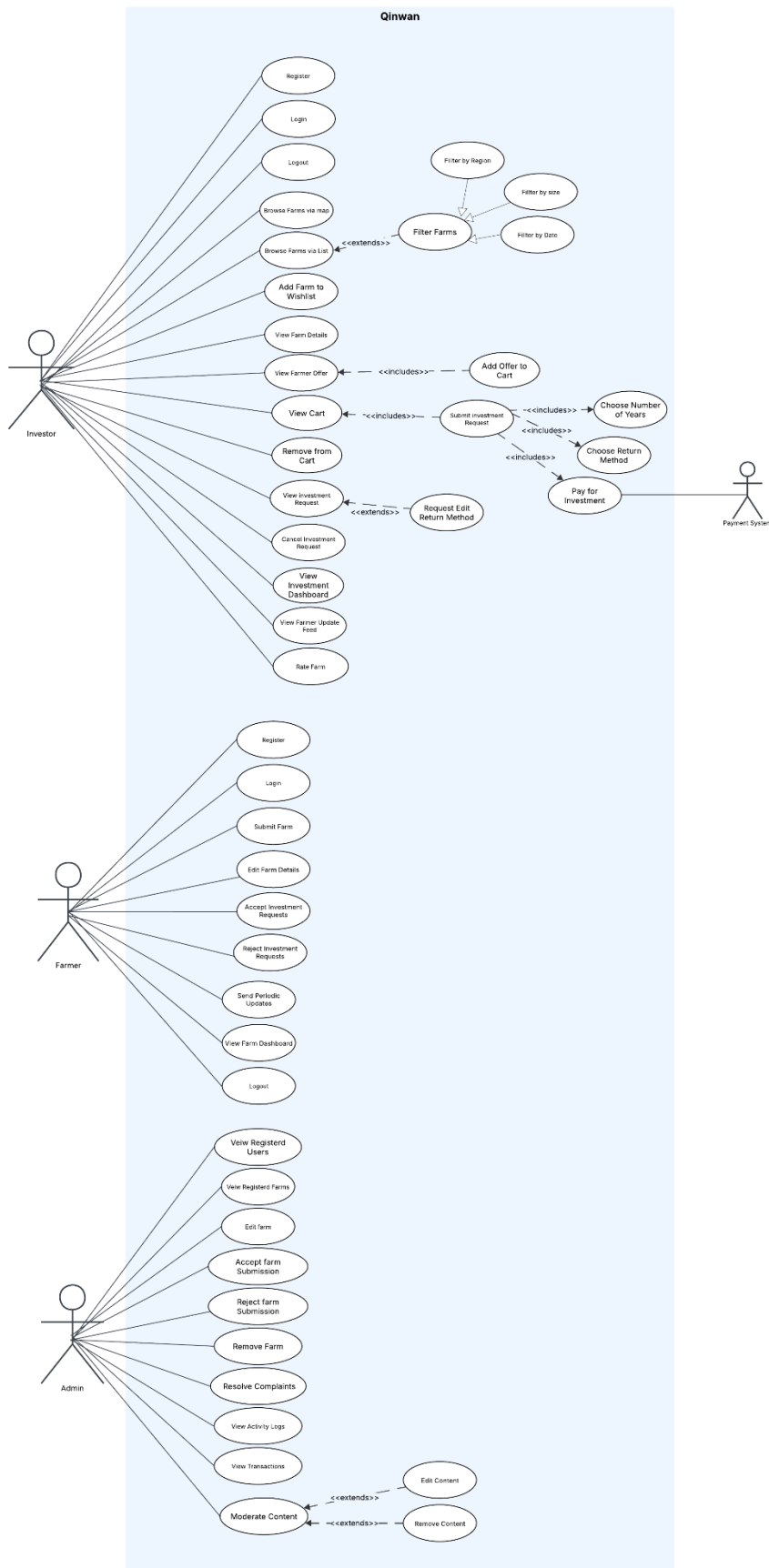


Figure 19 : Use Case Diagram - full diagram

## 3.4 Product Backlog

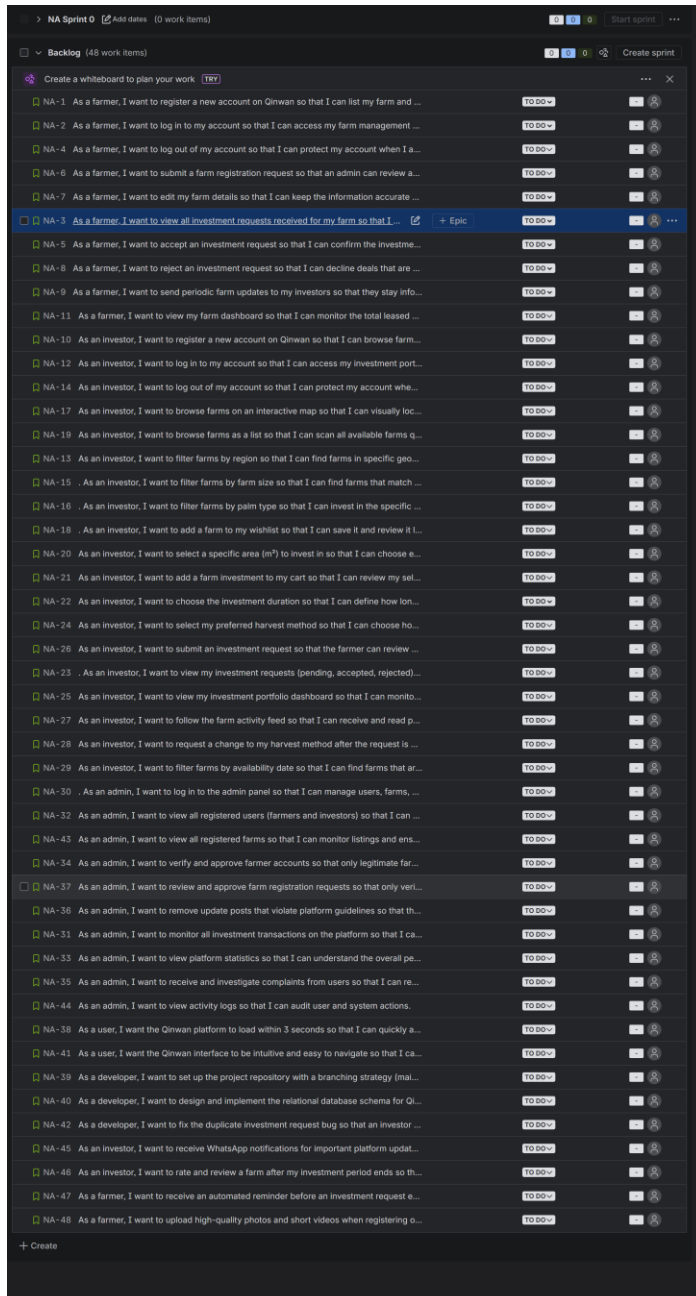


Figure 20 : Product Backlog

| PBI (User Story)                                                                                                           | Size (Story Points) | Type    | Acceptance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------|---------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Farmer</b>                                                                                                              |                     |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| US-01. As a farmer, I want to register a new account on Qinwan so that I can list my farm and receive investment requests. | 5                   | Feature | <ul style="list-style-type: none"> <li>The registration form must include: full name, national ID, phone number, email, and password.</li> <li>The username must be 4–20 characters (letters, numbers, underscores, or periods only).</li> <li>Email must follow the format <a href="#">xxxx@xxx.xxx</a>; duplicates are rejected with a clear error message.</li> <li>Password must contain at least 8 characters, one uppercase, one lowercase, one number, and one special character.</li> <li>Passwords must match in the confirm-password field before submission.</li> <li>The system must hash and salt passwords before storing them in the database.</li> <li>A confirmation email is sent to the farmer after successful registration.</li> <li>After registration, the farmer is automatically logged in and redirected to their dashboard.</li> </ul> |
| US-02. As a farmer, I want to log in to my account so that I can access my farm management dashboard securely.             | 3                   | Feature | <ul style="list-style-type: none"> <li>Login requires a valid registered email and correct password.</li> <li>If credentials are invalid, the system displays: 'Invalid email or password. Please try again.'</li> <li>After 5 consecutive failed attempts, the account is temporarily locked for 15 minutes.</li> <li>Upon successful login, the farmer is redirected to their personal dashboard.</li> <li>The system must use HTTPS for all login data transmission.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                |
| US-03. As a farmer, I want to log out of my account so that I can protect my account when I am done.                       | 1                   | Feature | <ul style="list-style-type: none"> <li>A 'Log Out' button must be visible and accessible from any page within the dashboard.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



|                                                                                                                                                |   |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                |   |         | <ul style="list-style-type: none"> <li>• The farmer is redirected to the login page after logout.</li> <li>• Accessing any protected page after logout must redirect the user to the login page.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| US-04. As a farmer, I want to submit a farm registration request so that an admin can review and approve my farm to appear on the platform.    | 8 | Feature | <ul style="list-style-type: none"> <li>• The farm submission form must include: farm name, location (city, region), total area (m<sup>2</sup>), type of palm, number of palm trees, expected yield (kg/season), and photos.</li> <li>• The farmer must be logged in before submitting a farm registration request.</li> <li>• All required fields must be validated before submission; missing fields display specific error messages.</li> <li>• After submission, the farm status is set to 'Pending Review'.</li> <li>• The farmer receives an notification confirming that the request has been submitted.</li> <li>• The system prevents duplicate farm registrations with the same name and location.</li> </ul> |
| US-05. As a farmer, I want to edit my farm details so that I can keep the information accurate and up to date.                                 | 5 | Feature | <ul style="list-style-type: none"> <li>• The farmer can edit: farm name, total area, expected yield, palm type, photos, and description.</li> <li>• The farmer must be authenticated and own the farm to make any edits.</li> <li>• Edited farms that change core details (area, location) must be re-reviewed and re-approved by the admin.</li> <li>• Changes are saved with a timestamp and logged for audit purposes.</li> <li>• The farmer receives a confirmation notification after successful edits.</li> </ul>                                                                                                                                                                                                |
| US-06. As a farmer, I want to view all investment requests received for my farm so that I can make informed acceptance or rejection decisions. | 5 | Feature | <ul style="list-style-type: none"> <li>• The farmer's dashboard must show a list of all investment requests with: investor name, requested area (m<sup>2</sup>), selected harvest method, and requested start date.</li> <li>• Requests must be filterable by status: Pending, Accepted, Rejected.</li> <li>• Each request must display the investor's verified profile badge if available.</li> </ul>                                                                                                                                                                                                                                                                                                                 |

|                                                                                                                                            |   |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------|---|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                            |   |         | <ul style="list-style-type: none"> <li>• The list must update in real-time or refresh upon page reload.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| US-07. As a farmer, I want to accept an investment request so that I can confirm the investment deal with the investor.                    | 5 | Feature | <ul style="list-style-type: none"> <li>• The farmer can only accept a request if the requested area does not exceed the remaining available area on the farm.</li> <li>• Upon acceptance, the system updates the farm's available area and marks the request as 'Accepted'.</li> <li>• The investor receives an immediate notification about the acceptance.</li> <li>• The system logs the acceptance with a timestamp.</li> <li>• The farmer cannot accept the same area from two different investors simultaneously.</li> </ul> |
| US-08. As a farmer, I want to reject an investment request so that I can decline deals that are not suitable.                              | 3 | Feature | <ul style="list-style-type: none"> <li>• The farmer can reject any pending request with an optional reason field.</li> <li>• Upon rejection, the request is marked as 'Rejected' and the investor is notified immediately.</li> <li>• Rejected requests are moved to a separate 'Rejected' section in the dashboard.</li> <li>• The farmer cannot reverse a rejection without creating a new manual arrangement.</li> </ul>                                                                                                        |
| US-09. As a farmer, I want to send periodic farm updates to my investors so that they stay informed about the farm's progress.             | 5 | Feature | <ul style="list-style-type: none"> <li>• The farmer can publish text and image updates from the farm management dashboard.</li> <li>• Updates are only delivered to investors who have an accepted investment in that specific farm.</li> <li>• Each update is timestamped and appears in the investor's activity feed.</li> <li>• The farmer can only send updates if they have at least one active investor.</li> <li>• The admin can remove any update that violates platform guidelines.</li> </ul>                            |
| US-10. As a farmer, I want to view my farm dashboard so that I can monitor the total leased area, number of investors, and expected yield. | 5 | Feature | <ul style="list-style-type: none"> <li>• The dashboard must display: total farm area (m<sup>2</sup>), total leased area (m<sup>2</sup>), remaining available area (m<sup>2</sup>), number of active investors, expected yield (kg/season), and estimated revenue.</li> </ul>                                                                                                                                                                                                                                                       |

|                                                                                                                                                         |   |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                         |   |         | <ul style="list-style-type: none"> <li>• Data must refresh automatically or upon page reload.</li> <li>• All numeric values must be accurate and consistent with the accepted investment records in the system.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| US-11 . As a farmer, I want to receive an automated reminder before an investment request expires so that I do not miss responding to pending requests. | 3 | Feature | <ul style="list-style-type: none"> <li>•The system sends a reminder notification to the farmer 24 hours before a pending investment request expires.</li> <li>•A second reminder is sent 2 hours before expiry if the farmer has not yet responded.</li> <li>•Reminders are delivered via WhatsApp and SMS.</li> <li>•If the farmer does not respond before the deadline, the request is automatically marked as 'Expired' and the investor is notified.</li> <li>•Expired requests are moved to a separate 'Expired' tab in the farmer's dashboard.</li> </ul>                                                                                                                                                                       |
| <b>Investor</b>                                                                                                                                         |   |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| US-01. As an investor, I want to register a new account on Qinwan so that I can browse farms and submit investment requests.                            | 5 | Feature | <ul style="list-style-type: none"> <li>• The registration form must include: full name, phone number, email, and password.</li> <li>• Email format must be validated (<a href="#">xxxx@xxx.xxx</a>); duplicate emails are rejected with a clear error message.</li> <li>• Password must contain at least 8 characters, one uppercase, one lowercase, one number, and one special character.</li> <li>• Passwords must match in the confirm-password field.</li> <li>• The system hashes and salts all passwords before storage.</li> <li>• A confirmation email is sent to the investor after registration.</li> <li>• After registration, the investor is automatically logged in and redirected to the home/browse page.</li> </ul> |
| US-02. As an investor, I want to log in to my account so that I can access my investment portfolio and browse farms.                                    | 3 | Feature | <ul style="list-style-type: none"> <li>• Login requires a valid registered email and correct password.</li> <li>• If credentials are invalid, the system displays a clear error message.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                                                                                                                                        |   |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                        |   |         | <ul style="list-style-type: none"> <li>• After 5 consecutive failed attempts the account is locked for 15 minutes.</li> <li>• Upon successful login, the investor is redirected to their personal dashboard.</li> </ul>                                                                                                                                                                                                                                                                     |
| US-03. As an investor, I want to log out of my account so that I can protect my account when I finish.                                                 | 1 | Feature | <ul style="list-style-type: none"> <li>• A 'Log Out' button must be visible and accessible from any page of the investor interface.</li> <li>• Upon logout, the session is terminated immediately and all session tokens are invalidated.</li> <li>• The investor is redirected to the login page.</li> </ul>                                                                                                                                                                               |
| US-04. As an investor, I want to browse farms on an interactive map so that I can visually locate farms across Saudi Arabia and explore their details. | 8 | Feature | <ul style="list-style-type: none"> <li>• The map must display all admin-approved farms as interactive pins.</li> <li>• Clicking a farm pin must display a summary card: farm name, region, total area, available area, palm type, and expected yield.</li> <li>• The map must be integrated with a national mapping system .</li> <li>• The map must load within 3 seconds for 90% of users.</li> <li>• If no farms exist in a region, no pins are shown and no error is thrown.</li> </ul> |
| US-05. As an investor, I want to browse farms as a list so that I can scan all available farms quickly without using the map.                          | 5 | Feature | <ul style="list-style-type: none"> <li>• The list view must show all approved farms with: name, region, available area, palm type, and expected yield.</li> <li>• Each list item must be clickable and redirect to the full farm detail page.</li> <li>• The list must reflect real-time availability of farm areas.</li> </ul>                                                                                                                                                             |
| US-06. As an investor, I want to filter farms by region so that I can find farms in specific geographic areas of interest.                             | 3 | Feature | <ul style="list-style-type: none"> <li>• The region filter must list all available Saudi regions/cities where farms exist.</li> <li>• Selecting a region updates both the map pins and the list view dynamically.</li> <li>• If no farms match the selected region, the system displays: 'No farms found in this region.'</li> <li>• The filter can be cleared with a single 'Clear Filters' button.</li> </ul>                                                                             |

|                                                                                                                                                    |   |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| US-07. As an investor, I want to filter farms by farm size so that I can find farms that match my investment capacity.                             | 3 | <b>Feature</b> | <ul style="list-style-type: none"> <li>• The size filter must offer predefined ranges (e.g., 0–500 m<sup>2</sup>, 500–2000 m<sup>2</sup>, 2000+ m<sup>2</sup>).</li> <li>• Applying the filter updates the results list and map dynamically.</li> <li>• If no farms match the size filter, the system displays: 'No farms match your selected size range.'</li> <li>• The filter can be cleared with a single 'Clear Filters' button.</li> </ul>                  |
| US-08. As an investor, I want to filter farms by palm type so that I can invest in the specific variety of dates I prefer.                         | 3 | <b>Feature</b> | <ul style="list-style-type: none"> <li>• The palm type filter must list all available date varieties registered in the system.</li> <li>• Applying the filter updates the results list and map dynamically.</li> <li>• Multiple palm types can be selected simultaneously.</li> <li>• If no farms match the selected type, the system displays: 'No farms match your selected palm type.'</li> </ul>                                                              |
| US-09. As an investor, I want to add a farm to my wishlist so that I can save it and review it later before making an investment decision.         | 3 | <b>Feature</b> | <ul style="list-style-type: none"> <li>• A 'Add to Wishlist' button must be visible on every farm card and farm detail page.</li> <li>• Clicking the button adds the farm to the investor's personal wishlist section.</li> <li>• Wishlist items persist across sessions (saved to the investor's account).</li> <li>• The investor can remove items from the wishlist at any time.</li> <li>• The wishlist is private and not visible to other users.</li> </ul> |
| US-10. As an investor, I want to select a specific area (m <sup>2</sup> ) to invest in so that I can choose exactly how much land I wish to lease. | 5 | <b>Feature</b> | <ul style="list-style-type: none"> <li>• The area selection input must only accept numeric values greater than zero.</li> <li>• The selected area must not exceed the farm's remaining available area; the system displays an error if exceeded.</li> <li>• The system must display the cost preview based on the selected area before the investor proceeds.</li> <li>• The investor cannot proceed to the cart without specifying a valid area.</li> </ul>      |

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| US-11. As an investor, I want to add a farm investment to my cart so that I can review my selections before submitting a request.    | 5 | Feature | <ul style="list-style-type: none"> <li>• The cart must display: farm name, selected area (m<sup>2</sup>), investment duration, selected harvest method, and estimated cost.</li> <li>• The investor can add multiple farms to the cart in a single session.</li> <li>• Items in the cart are preserved for up to 24 hours.</li> <li>• The investor can delete any item from the cart at any time.</li> </ul>                                                                                                     |
| US-12. As an investor, I want to choose the investment duration so that I can define how long I wish to lease the farm area.         | 3 | Feature | <ul style="list-style-type: none"> <li>• The investor must select a duration from predefined options (e.g., 1 season, 1 year, 2 years).</li> <li>• The selected duration must be shown clearly in the cart and on the submitted request.</li> <li>• The investor cannot submit an investment request without selecting a valid duration.</li> </ul>                                                                                                                                                              |
| US-13. As an investor, I want to select my preferred harvest method so that I can choose how I will benefit from the farm's produce. | 5 | Feature | <ul style="list-style-type: none"> <li>• The investor must choose one harvest method from: (1) receive dates delivered to their home, (2) sell the harvest and receive profits, or (3) donate the harvest to a charity.</li> <li>• The selected method must be displayed in the cart and on the investment request.</li> <li>• If the investor selects home delivery, a delivery address field becomes mandatory.</li> <li>• The investor cannot submit a request without selecting a harvest method.</li> </ul> |
| US-14. As an investor, I want to submit an investment request so that the farmer can review and accept my proposed investment.       | 8 | Feature | <ul style="list-style-type: none"> <li>• The investor must be logged in with a verified account to submit a request.</li> <li>• The cart must contain at least one valid item before submission is allowed.</li> <li>• Upon submission, the request is sent to the corresponding farmer and marked as 'Pending'.</li> <li>• The investor receives an on-platform and email notification confirming request submission.</li> </ul>                                                                                |

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|                                                                                                                                                        |   |         | <ul style="list-style-type: none"> <li>• The investor can view the submitted request in the 'Pending Requests' section of their dashboard.</li> <li>• If the cart is empty, the system shows: 'Please select a farm area before submitting a request.'</li> </ul>                                                                                                                                                                                                                               |
| US-15. As an investor, I want to view my investment requests (pending, accepted, rejected) so that I can track the status of all my submissions.       | 5 | Feature | <ul style="list-style-type: none"> <li>• The requests section must display three tabs: Pending, Accepted, Rejected.</li> <li>• Each request must show: farm name, requested area, duration, harvest method, submission date, and current status.</li> <li>• The investor can cancel a pending request at any time before the farmer responds.</li> <li>• Accepted and rejected requests are read-only and cannot be cancelled.</li> </ul>                                                       |
| US-16. As an investor, I want to view my investment portfolio dashboard so that I can monitor all my active investments at a glance.                   | 8 | Feature | <ul style="list-style-type: none"> <li>• The dashboard must display for each active investment: farm name, owner's contact, location on mini-map, leased area (m<sup>2</sup>), palm type in the region, expected yield, and estimated profit/loss summary.</li> <li>• The dashboard must show a total summary: total m<sup>2</sup> leased, total estimated yield, total estimated profit.</li> <li>• Data must be accurate and reflect the accepted investment records in real-time.</li> </ul> |
| US-17. As an investor, I want to follow the farm activity feed so that I can receive and read periodic updates sent by the farmers I have invested in. | 5 | Feature | <ul style="list-style-type: none"> <li>• The activity feed must display updates only from farms where the investor has an accepted investment.</li> <li>• Each update must show: farm name, update text, attached image (if any), and timestamp.</li> <li>• New updates must appear at the top of the feed.</li> <li>• The investor receives a push notification when a new update is posted by a farmer they are invested with.</li> </ul>                                                     |
| US-18. As an investor, I want to request a change to my harvest method after the request is accepted so that I can adjust how I receive my produce.    | 5 | Feature | <ul style="list-style-type: none"> <li>• This option is only available for investments that have already been accepted by the farmer.</li> <li>• The investor can submit a change request; the farmer must approve or reject the change.</li> </ul>                                                                                                                                                                                                                                             |

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|                                                                                                                                                 |   |         | <ul style="list-style-type: none"> <li>• The investor is notified of the farmer's decision via the platform and email.</li> <li>• Only one pending harvest-method change request is allowed per investment at a time.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                   |
| US-19. As an investor, I want to receive WhatsApp notifications for important platform updates so that I can stay informed .                    | 3 | Feature | <ul style="list-style-type: none"> <li>•The investor can enable WhatsApp notifications from their profile settings.</li> <li>•Notifications are sent for the following events: investment request accepted or rejected, new farm update posted, harvest method change request approved or rejected.</li> <li>•The investor must provide and verify a valid WhatsApp number during registration or from settings.</li> <li>•If WhatsApp delivery fails, the system falls back to SMS.</li> <li>•The investor can disable WhatsApp notifications at any time from their settings.</li> </ul>         |
| US-20. As an investor, I want to rate and review a farm after my investment period ends so that I can share my experience with other investors. | 3 | Feature | <ul style="list-style-type: none"> <li>•The rating option only appears after the investment period for that farm has officially ended.</li> <li>•The investor can submit a rating from 1 to 5 stars and an optional written review.</li> <li>•Each investor can only submit one rating per investment.</li> <li>•The rating is publicly visible on the farm's detail page.</li> <li>•The investor cannot edit or delete their rating after submission.</li> <li>•The farm's overall rating displayed on its card and detail page is calculated as the average of all submitted ratings.</li> </ul> |
| Admin                                                                                                                                           |   |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| US-01. As an admin, I want to view all registered users                                                                                         | 3 | Feature | <ul style="list-style-type: none"> <li>• The admin panel must display a list of all users with: name, role</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |



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| (farmers and investors) so that I can monitor platform membership.                                                                            |   |         | (farmer/investor), registration date, and status (active/suspended).<br>• The list must be searchable by name and filterable by role or status.<br>• The admin can click on any user to view their full profile and activity.                                                                                                                                                                                                                                                                                                                                   |
| US-02. As an admin, I want to verify and approve farmer accounts so that only legitimate farm owners operate on the platform.                 | 8 | Feature | <ul style="list-style-type: none"> <li>• The admin reviews the farmer's submitted national ID and farm ownership documents.</li> <li>• The admin can approve or reject a farmer account with a mandatory note.</li> <li>• Upon approval, the farmer receives a notification and can begin listing farms.</li> <li>• Upon rejection, the farmer receives a notification with the stated reason.</li> <li>• Approved farmers are marked with a verification badge on the platform.</li> </ul>                                                                     |
| US-03. As an admin, I want to review and approve farm registration requests so that only verified and accurate farms appear on the platform.  | 8 | Feature | <ul style="list-style-type: none"> <li>• The admin sees all pending farm registration requests with full details (name, location, area, photos, palm type).</li> <li>• The admin can approve the farm, which makes it visible to investors on the map and list.</li> <li>• The admin can reject the farm with a mandatory rejection reason sent to the farmer.</li> <li>• The admin can request additional information from the farmer before making a decision.</li> <li>• Approval and rejection actions are logged with a timestamp and admin ID.</li> </ul> |
| US-04. As an admin, I want to edit or remove farm listings that violate platform guidelines so that the platform maintains quality and trust. | 5 | Feature | <ul style="list-style-type: none"> <li>• The admin can edit any farm field (name, description, area, photos) at any time.</li> <li>• The admin can deactivate or permanently delete a farm with a mandatory reason.</li> <li>• Affected farmers are notified immediately upon any admin edit or deletion.</li> <li>• All admin edits and deletions are logged in the audit trail with a timestamp.</li> </ul>                                                                                                                                                   |

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| US-05. As an admin, I want to monitor all investment transactions on the platform so that I can ensure financial integrity and resolve issues.     | 8 | Feature | <ul style="list-style-type: none"> <li>• The admin panel must display all transactions with: investor name, farmer name, farm name, area, duration, harvest method, status, and date.</li> <li>• Transactions are filterable by status (pending, accepted, rejected, completed) and date range.</li> <li>• The admin can view the full detail of any individual transaction.</li> <li>• The admin receives an alert for any transactions flagged as suspicious.</li> </ul> |
| US-06. As an admin, I want to view platform statistics so that I can understand the overall performance and growth of Qinwan.                      | 5 | Feature | <ul style="list-style-type: none"> <li>• The admin dashboard must display: total registered users, total active farms, total leased area (m<sup>2</sup>), total investment transaction volume, and new registrations per month.</li> <li>• Statistics must update in real-time or at most with a 5-minute delay.</li> <li>• The admin can export statistics as an Excel or PDF report.</li> </ul>                                                                          |
| US-07. As an admin, I want to receive and investigate complaints from users so that I can resolve disputes fairly and maintain platform trust.     | 8 | Feature | <ul style="list-style-type: none"> <li>• Users (farmers and investors) can submit a complaint via a dedicated 'Report a Problem' form.</li> <li>• The admin sees all complaints in a dedicated complaints queue with: submitter, subject, description, and date.</li> <li>• The admin can change the complaint status to: Under Investigation, Resolved, or Dismissed.</li> <li>• Resolved complaints are archived and not deleted.</li> </ul>                             |
| US-08. As an admin, I want to remove update posts that violate platform guidelines so that the community content stays appropriate and respectful. | 3 | Feature | <ul style="list-style-type: none"> <li>• The admin can view all farmer-posted updates across all farms.</li> <li>• The admin can delete any update with a mandatory reason.</li> <li>• The farmer whose update was deleted is notified with the reason.</li> </ul>                                                                                                                                                                                                         |
| US-09. As an admin, I want to view all registered farms so that I can monitor listings and ensure platform compliance.                             | 3 | Feature | <ul style="list-style-type: none"> <li>• The admin panel displays a complete list of all registered farms.</li> </ul>                                                                                                                                                                                                                                                                                                                                                      |

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|                                                                                                                          |   |         | <ul style="list-style-type: none"> <li>•Each farm record shows: farm name, owner name, location, area (m<sup>2</sup>), palm type, and status.</li> <li>•The list is searchable by farm name or owner name.</li> <li>•The list is filterable by status (pending, approved, rejected, deactivated).</li> <li>•The admin can click on any farm to view full details.</li> </ul>                                                                                                                                                                                     |
| US-10. As an admin, I want to view activity logs so that I can audit user and system actions.                            | 5 | Feature | <ul style="list-style-type: none"> <li>•The admin can access a dedicated "Activity Logs" page.</li> <li>•The system displays logs including: user ID, action type, timestamp, and affected entity</li> <li>•Logs include actions such as: account approval, farm approval, edits, deletions, and transaction updates.</li> <li>•The logs are searchable by user name or action type.</li> <li>•The logs are filterable by date range.</li> <li>•Logs cannot be edited or deleted by any user.</li> <li>•The system records all actions in real-time.</li> </ul>  |
| US-11. As an admin, I want to edit user-generated content so that I can correct minor violations without deleting posts. | 5 | Feature | <ul style="list-style-type: none"> <li>•The admin can access a dedicated "Activity Logs" page.</li> <li>•The system displays logs including: user ID, action type, timestamp, and affected entity.</li> <li>•Logs include actions such as: account approval, farm approval, edits, deletions, and transaction updates.</li> <li>•The logs are searchable by user name or action type.</li> <li>•The logs are filterable by date range.</li> <li>•Logs cannot be edited or deleted by any user.</li> <li>•The system records all actions in real-time.</li> </ul> |
| <b>Non-Functional Requirements</b>                                                                                       |   |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

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| US-01. As a user, I want the Qinwan platform to load within 3 seconds so that I can quickly access my data without frustration.                                                                         | 5 | <b>Feature</b>        | <ul style="list-style-type: none"> <li>• The homepage and main browse page must load in under 3 seconds for 90% of users .</li> <li>• API response time must not exceed 1 second.</li> <li>• The interactive farm map must fully render within 3 seconds after page load.</li> </ul>                                                                                                                                                                                                                   |
| US-02. As a user, I want the Qinwan interface to be intuitive and easy to navigate so that I can complete key tasks without confusion.                                                                  | 5 | <b>Feature</b>        | <ul style="list-style-type: none"> <li>• A new user must be able to register and submit their first investment request within 5 minutes.</li> <li>• Key tasks (browse farms, submit request, view dashboard) must be completable in no more than 4 clicks from the home screen.</li> <li>• Error messages must be in clear Arabic and English, describing the problem and actionable steps to fix it.</li> </ul>                                                                                       |
| <b>Developer Stories</b>                                                                                                                                                                                |   |                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| US-01. As a developer, I want to set up the project repository with a branching strategy (main / develop / feature branches) so that the team can collaborate without overwriting each other's work.    | 3 | <b>Technical Work</b> | <ul style="list-style-type: none"> <li>• A Git repository is created and shared with all team members.</li> <li>• Branching conventions are documented: main (production), develop (staging), feature/xxx (per story).</li> <li>• Pull requests require at least one code review approval before merging into develop.</li> <li>• No direct commits are allowed to the main branch.</li> <li>• A README file documents the project setup steps and branching rules.</li> </ul>                         |
| US-02. As a developer, I want to design and implement the relational database schema for Qinwan so that all entities (users, farms, investments, transactions) are stored consistently and efficiently. | 8 | <b>Technical Work</b> | <ul style="list-style-type: none"> <li>• The database schema includes tables for: Users, Farmers, Investors, Admins, Farms, InvestmentRequests, Transactions, FarmUpdates, Complaints, and Wishlist.</li> <li>• All foreign key relationships are correctly defined and enforced at the database level.</li> <li>• All text fields storing sensitive data (passwords, IDs) use appropriate encrypted or hashed column types.</li> <li>• An Entity-Relationship Diagram (ERD) is documented.</li> </ul> |

| Defect Storie                                                                                                                                                |   |               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| US-01. As a developer, I want to fix the duplicate investment request bug so that an investor cannot submit more than one pending request for the same farm. | 3 | <b>Defect</b> | <ul style="list-style-type: none"> <li>• BUG: An investor can currently submit multiple pending requests for the same farm simultaneously, causing inconsistency in farm availability tracking.</li> <li>• FIX: Before allowing a new investment request submission, the system must check if the investor already has a 'Pending' or 'Accepted' request for the same farm.</li> <li>• If a duplicate is detected, the system displays: 'You already have an active request for this farm. Please wait for the farmer's response before submitting a new one.'</li> <li>• Verified by test: attempting to submit a second request for the same farm while the first is pending must be blocked.</li> </ul> |

Table 9 : Product Backlog

## 4 References

- [1] General Authority for Statistics (GASTAT), "Date production in the Kingdom reaches approximately 1.923 thousand tons in 2024," GASTAT, Nov. 30, 2025. [Online]. Available: <https://www.stats.gov.sa/en/w/news/127>. [Accessed: Feb. 11, 2026].
- [2] Food and Agriculture Organization of the United Nations, Date palm cultivation and production, 2021. [Online]. Available: <https://www.fao.org/home/en>. [Accessed: Feb. 11, 2026].
- [3] Haraj, "Haraj Online Marketplace," [Online]. Available: <https://haraj.com.sa>. [Accessed: Jan. 19, 2026].
- [4] Atyab Al-Tamor, "Official Website," [Online]. Available: <https://atayb.com>. [Accessed: Jan. 19, 2026].



## 5 Appendix

### 5.1 Appendix A : Interview

#### Interview questions

##### **For Investors :**

- Q1: Have you ever invested in agriculture before?
- Q2: Would you consider investing in date palm farms? Why?
- Q3: What information would you need before deciding to invest in a farm?
- Q4: What might make you stop using an agricultural investment platform?
- Q5: How would you prefer to track the progress of your investment?

##### **For Farmers :**

- Q1: Have you ever invested in agriculture before?
- Q2: Would you consider allowing others to invest in your date palm farm? Why?
- Q3: What information do you think should be shared with investors before they invest?
- Q4: What might make you stop using an agricultural investment platform?
- Q5: How would you prefer to share updates with investors?

## Interview Transcriptions

| Online Interview - farmer (1)                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interviewee:</b> Ibrahim                                                                                                                                                                                                                                                                                                                                                                                 | <b>Interviewer:</b> Jana                                                                                                                                                                                                         |
| <b>Location/Medium:</b> Ibrahim's house                                                                                                                                                                                                                                                                                                                                                                     | <b>Appointment Date:</b> 19-2-2026<br><b>Start Time:</b> 9:06 PM<br><b>End Time:</b> 9:21 PM                                                                                                                                     |
| <b>Objectives:</b> <ul style="list-style-type: none"> <li>Gain insight into investors' and farm owners' preferences in agricultural investment.</li> <li>Identify the key information required before making an investment decision.</li> <li>Understand usability challenges in agricultural investment platforms.</li> <li>Explore preferred methods for communication and investment updates.</li> </ul> | <b>Reminders:</b> <ul style="list-style-type: none"> <li>Has experience in agricultural investment and managing a date palm farm.</li> <li>Believes farming requires continuous supervision and personal involvement.</li> </ul> |
| <b>Agenda:</b><br>Introduction<br>Background on Project<br>Overview of Interview<br>Topics to Be Covered<br>Permission to Record<br>Question 1<br>Question 2<br>Question 3<br>Question 4<br>Question 5<br>Summary of Major Key Points<br>Questions from Interviewee<br>Closing                                                                                                                              | <b>Approximate Time:</b><br>1 min<br>3 min<br><br>1 min<br>30 sec<br>1.5 min<br>1 min<br>2 min<br>1 min<br>1 min<br>1 min<br>1 min<br>1 min<br>1 min                                                                             |
| <b>General Observations:</b> The interviewee appeared confident and experienced in agricultural investment. He provided clear and direct answers and showed practical knowledge about farm management and investment partnerships.                                                                                                                                                                          |                                                                                                                                                                                                                                  |
| <b>Topic not covered:</b> We did not discuss the preferred investment duration or the approval process for investor requests.                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                  |

Figure 21 : Interview Transcription - farmer 1



| Online Interview - farmer (1)                                                         |                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Interviewee: Ibrahim                                                                  | Date: 19-2-2026                                                                                                                                                                                                                                                                                                                                                                                    |
| Questions:                                                                            | Notes:                                                                                                                                                                                                                                                                                                                                                                                             |
| Q1: Have you ever invested in agriculture before?                                     | <p><b>Answer:</b><br/>Yes, I have previous experience in agricultural investment, either through direct farming or partnerships with other investors. However, it requires expertise and continuous monitoring.</p> <p><b>Observations:</b><br/>When talking about his experience, the interviewee showed that he understands agricultural investment needs regular follow-up and supervision.</p> |
| Q2: Would you consider allowing others to invest in your date palm farm? Why?         | <p><b>Answer:</b><br/>Yes, I would consider it because it helps reduce financial and operational costs.</p> <p><b>Observations:</b><br/>The interviewee appeared open to partnerships and sees them as a way to reduce costs.</p>                                                                                                                                                                  |
| Q3: What information do you think should be shared with investors before they invest? | <p><b>Answer:</b><br/>The available investment area , the source and availability of water, electricity infrastructure, the type of date palms planted, and the expected average annual yield.</p> <p><b>Observations:</b><br/>He focused on clear and basic farm information before investors make a decision.</p>                                                                                |
| Q4: What might make you stop using an agricultural investment platform?               | <p><b>Answer:</b><br/>If the platform is complicated or requires advanced technical skills.</p> <p><b>Observations:</b><br/>His answer shows that ease of use is important to him when using digital platforms.</p>                                                                                                                                                                                |
| Q5: How would you prefer to share updates with investors?                             | <p><b>Answer:</b><br/>I prefer sharing updates through periodic videos that show the condition of the farm and the date palms.</p> <p><b>Observations:</b><br/>He prefers sending video updates to show the farm's condition to investors.</p>                                                                                                                                                     |

Figure 22 : Interview Transcription - farmer1

| Online Interview - farmer (2)                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interviewee:</b> Saad                                                                                                                                                                                                                                                                                                                                                                                            | <b>Interviewer:</b> Noura Altuwiam                                                                                                                                                                                                     |
| <b>Location/Medium:</b> Saad's House                                                                                                                                                                                                                                                                                                                                                                                | <b>Appointment Date:</b> 23-2-2026<br><b>Start Time:</b> 4:10 PM<br><b>End Time:</b> 4:24 PM                                                                                                                                           |
| <b>Objectives:</b> <ul style="list-style-type: none"> <li>• Gain insight into investors' and farm owners' preferences in agricultural investment.</li> <li>• Identify the key information required before making an investment decision.</li> <li>• Understand usability challenges in agricultural investment platforms.</li> <li>• Explore preferred methods for communication and investment updates.</li> </ul> | <b>Reminders:</b> <ul style="list-style-type: none"> <li>• Has several years of experience working in agriculture.</li> <li>• Is generally interested in developing agricultural practices and improving farm productivity.</li> </ul> |
| <b>Agenda:</b><br>Introduction<br>Background on Project<br>Overview of Interview<br>Topics to Be Covered<br>Permission to Record<br>Question 1<br>Question 2<br>Question 3<br>Question 4<br>Question 5<br>Summary of Major Key Points<br>Questions from Interviewee<br>Closing                                                                                                                                      | <b>Approximate Time:</b><br>1 min<br>3 min<br>1 min<br>30 sec<br>1 min<br>1.5 min<br>2 min<br>1 min<br>1 min<br>1 min<br>1 min<br>1 min                                                                                                |
| <b>General Observations:</b> During the interview, the interviewee was confident and engaged. the interviewee appeared confident and engaged. He showed clear experience in agriculture and expressed support for involving investors to expand his farm.                                                                                                                                                           |                                                                                                                                                                                                                                        |
| <b>Topic not covered:</b> We did not discuss the investment duration, the approval process for investor requests, or how profits would be distributed.                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                        |

Figure 23 : Interview Transcription - farmer2

| Online Interview - farmer (2)                                                         |                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interviewee: Saad                                                                     | Date: 23-2-2026                                                                                                                                                                                                                                                                                                    |
| Questions:                                                                            | Notes:                                                                                                                                                                                                                                                                                                             |
| Q1: Have you ever invested in agriculture before?                                     | <p><b>Answer:</b><br/>Yes, I work in the agricultural field and have experience in farm management and production.</p> <p><b>Observations:</b><br/>The interviewee has experience in agriculture and shows confidence in farm management and production.</p>                                                       |
| Q2: Would you consider allowing others to invest in your date palm farm? Why?         | <p><b>Answer:</b><br/>Yes, to expand and increase production without high financial pressure and to improve the quality of agricultural services by providing additional liquidity.</p> <p><b>Observations:</b><br/>He supports allowing investors as a way to expand production and improve farm performance.</p> |
| Q3: What information do you think should be shared with investors before they invest? | <p><b>Answer:</b><br/>The number of palm trees, expected production, operational costs, the type of dates grown, and the water source.</p> <p><b>Observations:</b></p>                                                                                                                                             |
| Q4: What might make you stop using an agricultural investment platform?               | <p><b>Answer:</b><br/>If the platform is complicated to use or if there are high commission fees.</p> <p><b>Observations:</b><br/>He is concerned about platform usability and commission costs</p>                                                                                                                |
| Q5: How would you prefer to share updates with investors?                             | <p><b>Answer:</b><br/>Through seasonal reports, along with periodic photos and videos.</p> <p><b>Observations:</b><br/>He prefers communication through reports and visual updates to maintain investor trust.</p>                                                                                                 |

Figure 24 : Interview Transcription - farmer2

| Online Interview - investor (1)                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interviewee:</b> Mohammed                                                                                                                                                                                                                                                                                                                                                                                        | <b>Interviewer:</b> Haya                                                                                                                                                                                                     |
| <b>Location/Medium:</b> Mohmmmed's house                                                                                                                                                                                                                                                                                                                                                                            | <b>Appointment Date:</b> 20-2-2026<br><b>Start Time:</b> 10:20 PM<br><b>End Time:</b> 10:36 PM                                                                                                                               |
| <b>Objectives:</b> <ul style="list-style-type: none"> <li>• Gain insight into investors' and farm owners' preferences in agricultural investment.</li> <li>• Identify the key information required before making an investment decision.</li> <li>• Understand usability challenges in agricultural investment platforms.</li> <li>• Explore preferred methods for communication and investment updates.</li> </ul> | <b>Reminders:</b> <ul style="list-style-type: none"> <li>• Has prior experience in agricultural investment.</li> <li>• Believes agricultural investment requires continuous supervision and hands-on involvement.</li> </ul> |
| <b>Agenda:</b><br>Introduction<br>Background on Project<br>Overview of Interview<br>Topics to Be Covered<br>Permission to Record<br>Question 1<br>Question 2<br>Question 3<br>Question 4<br>Question 5<br>Summary of Major Key Points<br>Questions from Interviewee<br>Closing                                                                                                                                      | <b>Approximate Time:</b><br>1 min<br>3 min<br>1 min<br>30 sec<br>1 min<br>1.5 min<br>2 min<br>1 min<br>1 min<br>1 min<br>2 min<br>1 min                                                                                      |
| <b>General Observations:</b> The interviewee appeared confident and experienced in agricultural investment. He seemed attentive and engaged throughout the discussion.                                                                                                                                                                                                                                              |                                                                                                                                                                                                                              |
| <b>Topic not covered:</b> We did not discuss detailed profit expectations or investment duration options.                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                              |

Figure 25 : Interview Transcription - Investor1

| Online Interview - investor (1)                                          |                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interviewee: Mohammed                                                    | Date: 20-2-2026                                                                                                                                                                                                                                                                                                                                        |
| Questions:                                                               | Notes:                                                                                                                                                                                                                                                                                                                                                 |
| Q1: Have you ever invested in agriculture before?                        | <p><b>Answer:</b><br/>Yes, I have previously invested in the agricultural sector. It was a valuable experience, but it requires continuous follow-up and supervision.</p> <p><b>Observations:</b><br/>When talking about his experience, the interviewee showed that he understands agricultural investment needs regular follow-up and attention.</p> |
| Q2: Would you consider investing in date palm farms? Why?                | <p><b>Answer:</b><br/>Perhaps, especially if I do not have suitable land available or if I do not have enough time to manage it myself.</p> <p><b>Observations:</b><br/>The interviewee decision depends on his personal situation, such as land availability and time.</p>                                                                            |
| Q3: What information would you need before deciding to invest in a farm? | <p><b>Answer:</b><br/>I would need to know about water availability and quality, the type and fertility of the soil, and the type of date palms planted.</p> <p><b>Observations:</b><br/>He focused on basic farm information such as water, soil, and palm type before investing.</p>                                                                 |
| Q4: What might make you stop using an agricultural investment platform?  | <p><b>Answer:</b><br/>Limited technical experience due to age could make using the platform difficult, especially if the platform is unclear or complicated to use.</p> <p><b>Observations:</b><br/>His answer suggests that ease of use is important for him when using digital platforms.</p>                                                        |
| Q5: How would you prefer to track the progress of your investment?       | <p><b>Answer:</b><br/>I prefer to track my investment through direct visits to the farm.</p> <p><b>Observations:</b><br/>He prefers visiting the farm in person to follow up on his investment.</p>                                                                                                                                                    |

Figure 26 :Interview Transcription - Investor 1

| Online Interview - investor (2)                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interviewee:</b> Abdullah                                                                                                                                                                                                                                                                                                                                                                                        | <b>Interviewer:</b> Noora Aluqaili                                                                                                                                                                        |
| <b>Location/Medium:</b> Noora's house                                                                                                                                                                                                                                                                                                                                                                               | <b>Appointment Date:</b> 21-2-2026<br><b>Start Time:</b> 7:21 PM<br><b>End Time:</b> 7:36:30 PM                                                                                                           |
| <b>Objectives:</b> <ul style="list-style-type: none"> <li>• Gain insight into investors' and farm owners' preferences in agricultural investment.</li> <li>• Identify the key information required before making an investment decision.</li> <li>• Understand usability challenges in agricultural investment platforms.</li> <li>• Explore preferred methods for communication and investment updates.</li> </ul> | <b>Reminders:</b> <ul style="list-style-type: none"> <li>• Has no prior experience in agricultural investment</li> <li>• Prefers investment opportunities that do not require daily management</li> </ul> |
| <b>Agenda:</b><br>Introduction<br>Background on Project<br>Overview of Interview<br>Topics to Be Covered<br>Permission to Record<br>Question 1<br>Question 2<br>Question 3<br>Question 4<br>Question 5<br>Summary of Major Key Points<br>Questions from Interviewee<br>Closing                                                                                                                                      | <b>Approximate Time:</b><br>1 min<br>3 min<br>1 min<br>30 sec<br>1 min<br>1.5 min<br>2 min<br>1 min<br>1 min<br>1 min<br>1 min<br>1.5 min<br>1 min                                                        |
| <b>General Observations:</b> During the interview, the interviewee showed interest in agricultural investment and was engaged in answering the questions. Although he has no prior experience, he appeared motivated to explore this opportunity.                                                                                                                                                                   |                                                                                                                                                                                                           |
| <b>Topic not covered:</b> We did not discuss detailed investment duration options, or how often returns would be distributed.                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                           |

Figure 27 : Interview Transcription - Investor2

| Online Interview - investor (2)                                          |                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interviewee: Abdullah                                                    | Date: 21-2-2026                                                                                                                                                                                                                                                                                      |
| Questions:                                                               | Notes:                                                                                                                                                                                                                                                                                               |
| Q1: Have you ever invested in agriculture before?                        | <p><b>Answer:</b><br/>No, I have not invested in this sector before, but I am interested in trying if the opportunity is available.</p> <p><b>Observations:</b><br/>The interviewee has no prior experience but shows interest in trying agricultural investment if an opportunity is available.</p> |
| Q2: Would you consider investing in date palm farms? Why?                | <p><b>Answer:</b><br/>Yes, definitely, because I do not need prior experience, and I can invest without owning land or managing it on a daily basis.</p> <p><b>Observations:</b><br/>He is strongly interested in investing because it does not require land ownership or daily management.</p>      |
| Q3: What information would you need before deciding to invest in a farm? | <p><b>Answer:</b><br/>The expected profit percentage, irrigation quantity, and the types of natural and chemical fertilizers used.</p> <p><b>Observations:</b><br/>He focuses on financial returns and technical farm details before making an investment decision.</p>                              |
| Q4: What might make you stop using an agricultural investment platform?  | <p><b>Answer:</b><br/>If the platform is unclear or complicated, and if there is no direct way to communicate with the farmer.</p> <p><b>Observations:</b></p>                                                                                                                                       |
| Q5: How would you prefer to track the progress of your investment?       | <p><b>Answer:</b><br/>I prefer to track my investment through periodic photos.</p> <p><b>Observations:</b><br/>He prefers visual updates, such as photos, to follow up on his investment.</p>                                                                                                        |

Figure 28 : Interview Transcription - Investor2

## 3.5 Appendix B : Questionnaires

### Questionnaires Link:

#### Investor:

[https://docs.google.com/forms/d/e/1FAIpQLSdf27j6CtEKrULC9ejLB0Qpwx\\_Z9jLT2NHHmuAN5GiihY-rdg/viewform?usp=publish-editor](https://docs.google.com/forms/d/e/1FAIpQLSdf27j6CtEKrULC9ejLB0Qpwx_Z9jLT2NHHmuAN5GiihY-rdg/viewform?usp=publish-editor)

#### Farmer:

<https://docs.google.com/forms/d/e/1FAIpQLSc7D0rWVb1K9buMA6EYLL8EK7r2AigCylTaJr54Cf6qkfutag/viewform?usp=header>

### Questionnaires Responses:

#### Investor:

[https://docs.google.com/spreadsheets/d/18hj\\_dvp551h4Bj9RGSzQs9diXMomYS66Vnh6eO2XCJA/edit?usp=sharing](https://docs.google.com/spreadsheets/d/18hj_dvp551h4Bj9RGSzQs9diXMomYS66Vnh6eO2XCJA/edit?usp=sharing)

#### Farmer:

<https://docs.google.com/spreadsheets/d/1EUKt8gLtJJ3fGuTIBZ6riWcVVBd5DgYzHmNbTks.JrAo/edit?usp=sharing>

### Questionnaires :

#### Investor:

Q1. ما هي فنتك العمرية ؟

- أقل من 25
- سنة 25-35
- سنة 36-45
- سنة فأكثر 46



Q2. عند استعراض المزارع، أي طريقة تفضل أكثر؟

- قائمة مع تفاصيل (List View)
- كلاهما متاح وأختار حسب الحاجة
- لا تفرق

Q3. ما هي معايير التصفية الأهم لك عند البحث عن مزرعة للاستثمار؟

- المنطقة الجغرافية
- (حجم المزرعة م<sup>2</sup>)
- نوع التمر / النخيل
- تقييم المزرعة

Q4. ما مدى أهمية عرض صور عالية الجودة للمزرعة في صفحة التفاصيل قبل اتخاذ قرار الاستثمار؟

(مقياس من 1 إلى 5)

1 2 3 4 5  
غير مهم ضروري

Q5. ما نوع تحديثات المزرعة الذي تريد تلقيه من صاحب المزرعة؟

- نص مكتوب فقط
- صور فقط
- مقاطع فيديو قصيرة
- نص + صور معا

Q6. كيف تفضل أن تصلك إشعارات المنصة؟

- رسالة عبر البريد الإلكتروني
- (SMS) عن طريق الرسائل النصية
- عن طريق الواتساب
- كل ما سبق

Q7. ما هي طريقة الاستفادة من الحصاد التي تفضلها؟

- استلام التمور بالتوصيل المنزلي
- التبرع بالحصاد لجمعية خيرية
- بيع الحصاد واستلام الأرباح
- تفضيل تحديد الطريقة لاحقاً

Q8. هل هناك ميزة أو خدمة تتمنى وجودها في منصة قنوان لم تُذكر في الأسئلة السابقة؟

(سؤال مفتوح)

**Farmar:**

**Q1. هل سبق لك تأجير جزء من مزرعتك للغير؟**

- نعم
- لا، لكنني مهتم
- لا، ولا أنوي

**Q2. عند استقبال طلب استثمار، ما المعلومات التي تريد رؤيتها عن المستثمر؟**

- الاسم الكامل والهوية الوطنية
- المساحة المطلوبة ومدة الاستثمار
- رقم الجوال للتواصل المباشر
- طريقة الاستفادة من الحصاد المختارة
- تاريخ الطلبات السابقة على المنصة

**Q3. ما المدة المناسبة لك للرد على طلب استثمار قبل انتهاء صلاحيته؟**

- ساعة 24
- ساعة 48
- أسبوع كامل
- بدون قيد زمني

**Q4. ما نوع المحتوى الذي تجد نفسك ستشاركه مع مستثمريك كتحديث دوري؟**

- (نص مكتوب فقط) وصف الوضع الراهن
- صور للمزرعة والمحصول
- مقاطع فيديو قصيرة
- صور + نص معا
- لا أنوي إرسال تحديثات

**Q5. كم مرة تريد إرسال تحديثات للمستثمرين؟**

- أسبوعيا
- كل أسبوعين
- شهريا
- (...عند وجود حدث مهم فقط (نمو، حصاد، مشكلة
- يكفي تحديث واحد في الموسم

**Q6. كيف تفضل أن تصلك إشعارات المنصة؟**

- رسالة بريد إلكتروني (Email)
- رسالة نصية (SMS)
- واتساب (WhatsApp)
- كل ما سبق للأحداث العاجلة

Q7. ما المعلومات التي تريد رؤيتها في لوحة التحكم الخاصة بك كمالك مزرعة؟

- إجمالي الأرض المؤجرة ونسبتها من المجموع
- تاريخ آخر تحديث أرسلته
- عدد المستثمرين النشطين
- إجمالي الأرباح التقديرية
- قائمة بأسماء المستثمرين وطلباتهم

Q8. ما أكبر تحدٍ تواجهه حالياً في تأجير أرضك الزراعية؟ وهل تعتقد أن منصة قنوان ستساعدك في حله؟

(سؤال مفتوح)